Labor Market Deregulation and Deaths of Despair: Evidence from Advanced Economies

Nara Chung*

October 13, 2026

Abstract

Employment protection legislation (EPL) deregulation has been implemented across advanced economies since 1980s. Although some studies looked at its impact on labor markets, a more downstream effect on population health such as deaths of despair was not explored. This paper asks whether such deregulation contributes to deaths of despair (drug overdoses, alcohol-related disease, and suicide), a mortality pattern first highlighted in the United States. Using OECD employment protection indicators and WHO mortality data for 26 economies from 1998–2019, I apply local projection method to trace the dynamic mortality response to major EPL reforms. Drug use disorder and suicide mortality rise significantly within five years of deregulation, with no significant rise in alcohol-related mortality. Some of the mortality responses are moderated by institutional context: health-system effectiveness attenuates drug and inflammatory heart diseases mortality; income support does so against suicide and violence mortality but ironically amplifies the alcohol-mortality response; and a higher poverty dampens the drug-mortality response. As suggestive evidence of possible mechanisms of material strain and labor market disconnection, net family income for a low-wage family decreases and short-tenure employment share increases significantly following a major EPL deregulation. These findings indicate that labor market deregulation carries public-health costs which can be mitigated by social safety nets and health-system capacity.

Keywords: Deaths of Despair; Employment Protection Legislation; Labor Market Deregulation; Mortality; Social Safety Nets; OECD Economies
JEL Classification Numbers: I14; J65; J08; I18

*Colorado State University, Email: nara.chung@colostate.edu

1 Introduction

Over the past three decades, advanced economies have experienced structural changes in labor markets and the distribution of economic gains (Ciminelli et al., 2022). A prominent trend is the weakening of workers' bargaining power and declining labor share of national income, which can be attributed to deregulation of employment protection legislation (EPL), declining union density, and the rise of non-standard work. On the other hand, Case and Deaton (2022) and Case and Deaton (2015, 2020) have shed light on the issue of "deaths of despair": mortality from drug overdoses, alcohol-related diseases, and suicide, which have led to reversed improvements in life expectancy in U.S. A number of studies found that this trend of deaths of despair is not just behavioral disorder of individuals but has been induced based on structural changes in the society (Beseran et al., 2022; Bjorklund, 2023; Geronimus et al., 2019; Glei et al., 2024; Hawkins et al., 2021; Kuo & Kawachi, 2023; Monnat, 2020; Scheiring et al., 2020). Although this phenomenon of deaths of despair has not emerged clearly in other advanced economies, these developments raise an question: do labor-market institutions shaped by EPL deregulation contribute to despair-related mortality among OECD countries?

This paper explores that question in a cross-country setting by examining the dynamic effects of major EPL reforms over 5-year horizon on cause-specific mortality. I construct annual data for 26 OECD economies from 1998 to 2019 on age-standardized death rates for drug use disorders, alcohol use disorders, and suicide as primary outcomes of deaths of despair. I also investigate cirrhosis of the liver, violence, and inflammatory heart disease mortality, which are plausibly associated with the same substance-use and psychosocial-stress pathways that drive the primary outcomes. To trace the timing and persistence of effects, we use local projections (Jordà, 2005) method to estimate impulse-response functions (IRFs) following major EPL reforms. This approach can characterize how deaths of despair respond in the years immediately after reforms and to project medium-run trajectories across causes of death. Among different versions of OECD's EPL indicator, a version that regards individual and collective dismissals on regular contracts is used for a corresponding time coverage with mortality data. I include both tightening and liberalizing reforms which is interchangeably called deregulation and identify major reforms from large discrete movements in the OECD's EPL indicator. In a panel framework, a baseline specification inlcudes country and year fixed effects, forward-looking reform controls, and lags of outcomes and treatments to mitigate reverse causality. This design is appropriate to policy shocks that occur at different times across countries. I then extend the baseline specification with interaction terms that capture how country characteristics (unemployment-benefit replacement rates, public social-transfer expenditure, public health-insurance coverage, avoidable mortality, and poverty) amplify or dampen the mortality response.

Results show three main findings. First, a major EPL reform is followed by a statistically and economically significant rise in drug use disorder mortality with effects that continue to increase over 5-year horizon. Suicide mortality exhibits a positive and significant increase within five years of a major reform. Second, the average effect on alcohol use disorder mortality is not statistically different from zero, pointing to heterogeneous causal-mechanisms across the components of deaths of despair. Third, institutional moderation of these effects is identified even though it is not uniform across every institution and cause of death. Countries with lower avoidable mortality and higher public health insurance coverage reveal significant decreases in drug while only higher public health insurance coverage is led to a decrease in inflammatory heart disease mortality, compared to countries with higher avoidable mortality and lower public health insurance coverage. Higher public social-transfer expenditure and unemployment-benefit replacement rates reduce suicide and violence mortality respectively in the comparison between 25th percentile and 75th percentile in those characteristics. Interestingly, higher public social-transfer expenditure is associated with significantly larger mortality in alcohol use disorder. By differentiating

the impacts of tightening and liberalizing reforms in EPL, it is demonstrated that the mortality responses of deregulation and tightening reforms are not symmetrical in the opposite direction with a formal test rejecting exact symmetry. These patterns are discussed further with possible mechanisms and on the account of institutional buffering in Section 6.

These results fill gaps in the literature. In labor economics, although some works assess the efficiency and distributional effects of EPL reforms (Barbieri & Cutuli, 2016; Ciminelli et al., 2022), yet few studies follow their downstream health outcomes. By connecting major EPL changes to cause-specific mortality dynamics, the current study provides new evidence that labor market deregulation carries measurable population-health costs. In population health and social epidemiology, this paper contributes to the deaths-of-despair literature developed mostly around U.S. within-country evidence (Case & Deaton, 2022; Case & Deaton, 2015, 2020; Gaydosh et al., 2019; Monnat, 2020; Ruhm, 2022; Ruhm, 2018; Stein et al., 2017) by adding cross-country evidence. In addtion, it complements individual-level research which connects insecurity, job loss, and precarious work to mental health, substance use, and suicide (Glei et al., 2024; Hawkins et al., 2021; Jolivet & Postel-Vinay, 2025; Scheiring et al., 2020) by illustrating that macro-institutional shifts can move despair-related mortality at the country level. Finally, by disaggregating deaths of despair and emphasizing timing, magnitude, and moderators, this study can allude to whether deaths of despair reflects just supply-side shocks (e.g., synthetic opioids) or deeper structural distress (Case & Deaton, 2022; Ruhm, 2022; Ruhm, 2018): the results of this paper are more inclined to a view that institutional erosion raises exposure and vulnerability to despair-related mortality even if drug supply conditions can amplify more acute and immediate harm.

The findings have the following policy implications. Since labor market reforms can increase drug and suicide mortality, labor-market institutions should be considered part of public-health agendas. The moderation observed in countries with stronger income supports and health systems advocates that policymakers contemplating major EPL reforms should accompany them with resilient social insurance and health-system capacity. In other words, the evidence cautions against narrow assessment of labor-market reform focusing on employment or wage while overlooking externalities in public health.

The remainder of the paper proceeds as the following. Section 2 discusses the related literature and develops the conceptual mechanisms linking deregulation to mortality. Section 3 describes the data sources, construction of outcome variables, and identification of major EPL reforms. Section 4 delineates empirical methodology of local projections and specifications. Section 5 reports the main results with the moderators' estimates along with supplementary analyses of candidate mediators and the asymmetry between deregulation and tightening. Section 6 discusses mechanisms, alternative explanations, institutional buffering, policy implications, and limitations. Section 7 concludes this paper.

2 Related Literature and Conceptual Motivation

This section reviews the three literatures that motivate our analysis, states our contribution relative to each, and develops the conceptual mechanisms linking employment protection (EPL) deregulation to deaths of despair that guide our empirical design.

2.1 Literature and Contribution

A first literature evaluates the labor-market consequences of employment protection legislation (EPL) deregulation, a central feature of labor market reform in advanced economies since the 1980s. Ciminelli et al. (2022) show, using a panel of 26 advanced economies from 1970–2013, that EPL deregulation significantly reduces the labor share of income, accounting for roughly 15% of its average decline. Barbieri and Cutuli (2016) find that partial, "marginal" EPL deregulation in Europe which relaxes protections

for temporary contracts while leaving permanent employment largely untouched widens labor market dualism and produces at best short-lived employment gains, with temporary jobs rarely serving as stepping stones to stable employment. This erosion of job protection is one part of a broader decline in labor power across advanced economies, evident also in the sharp fall in union density and its contribution to rising wage inequality (Peters, 2008; Western & Rosenfeld, 2011). This literature establishes that EPL deregulation carries real efficiency and distributional costs, but it has focused almost exclusively on labor-market outcomes; its downstream consequences for population health remain largely unexamined.

A second literature documents the rise of "deaths of despair" which is mortality from drug overdose, alcohol-related disease, and suicide and traces it to economic and social decline. Case and Deaton (2020) and Case and Deaton (2022) link the U.S. rise in these deaths, concentrated among adults without a bachelor's degree (U.S. Census Bureau, 2025), to stagnant wages, weakening attachment to stable employment, and eroding social institutions. Subsequent work extends this account along several dimensions: structural life-cycle evidence on the joint dynamics of job loss and mental health (Jolivet & Postel-Vinay, 2025); deindustrialization as a slow-acting driver of psychosocial stress and eroded social identity (Scheiring et al., 2020); occupation-specific exposure to injury, insecurity, and non-standard work (Hawkins et al., 2021); the interaction of economic precarity with opioid availability and community composition (Bjorklund, 2023); the compounding effects of income inequality and low social mobility (Kuo & Kawachi, 2023); causal evidence linking housing instability to substance-related mortality (Bradford & Bradford, 2020); direct evidence that labor-market policies such as minimum wages and the EITC can reduce specific despair-related deaths (Dow et al., 2020); causal and correlational evidence linking housing wealth and housing policy to despair-related mortality (Jou et al., 2023; Venkataramani & Tsai, 2020); financial losses as a mediating pathway between education and despair-associated risk behaviors (Fishman & Gutin, 2021); occupation- and employment-status-specific mortality risk together with life-course evidence connecting labor-market conditions to mental health (Gutin & Hummer, 2020; Hammarström et al., 2024); structural drivers of widening life-expectancy inequality (Geronimus et al., 2019); and a broader scoping review tying these deaths to widening social and economic inequality (Beseran et al., 2022). Reassessing this evidence, Glei et al. (2024) find that structural detachment from stable work and social ties predicts deaths of despair more robustly than subjective distress per se, reinforcing the view that these deaths reflect social and economic disconnection rather than purely psychological despair. This literature is overwhelmingly correlational and drawn from single-country, typically U.S., settings; it has not exploited a plausibly exogenous, cross-national policy shock to identify the labor-market channel directly.

A third literature debates whether the rise in these deaths mainly reflects changes in the drug-supply environment or demand rooted in economic distress. Ruhm (2018) argues that changes in the drug environment including the substitution from prescription opioids to heroin and, later, illicit fentanyl account for most of the rise in drug mortality, finding that economic decline explains only a small share. Ruhm (2022) shows that the post-1999 rise in midlife mortality is predominantly drug-overdose driven, with distinct age- and cause-specific patterns for suicide and alcohol-related disease, cautioning against treating deaths of despair as a single, homogeneous phenomenon. Reconciling these views, Case and Deaton (2022) argues that supply and demand interact rather than compete: the opioid epidemic would not have reached its scale absent underlying economic distress, even as the timing and severity of the crisis reflect drug-supply dynamics. This debate motivates two choices in our empirical design: we model each cause of death separately rather than pooling them, and we test whether institutional context systematically moderates the mortality response.

This paper's contribution is to bring these three literatures together within a single, plausibly exogenous research design. Relative to the labor economics literature, we extend the study of EPL deregulation to a health-related margin rarely part of the standard cost-benefit calculus. Relative to the deaths-of-

despair literature, we move beyond single-country, correlational evidence by tracing the dynamic mortality response to a discrete, cross-national institutional shock across 26 economies over more than two decades. Relative to the supply-side/demand-side debate, we provide new evidence on how the mortality response varies with health-system effectiveness, income-support generosity, and poverty, offering a way to assess whether labor-market institutions shape exposure to this risk independent of local drug-supply conditions. Our design also parallels a small set of studies that use quasi-experimental economic shocks to trace mortality effects, such as trade liberalization (Pierce & Schott, 2020) and automation (Mishra et al., 2024), but applies this approach to an institutional rather than a technological or trade-driven shock, across a cross-national panel of advanced economies.

2.2 Conceptual Motivation

Given this literature and the gap it leaves open, what should we expect when a country undertakes major EPL deregulation? We organize our expectations around three mechanisms linking deregulation to deaths of despair.

1. Economic Insecurity and Material Strain Deregulation increases job instability, unemployment risk, and income volatility. These forms of economic insecurity are established drivers of adverse health outcomes (Benach et al., 2014; Knapp et al., 2019; Link & Phelan, 1995; Wilkinson & Marmot, 2003), operating through financial strain, reduced future prospects, and heightened uncertainty, and increase the likelihood of risky or self-destructive behaviors.

2. Labor Market Disconnection and Psychosocial Stress Weaker job protection increases precarious employment and detachment from stable work, eroding social integration and undermining identity and purpose (Jolivet & Postel-Vinay, 2025; Putnam, 2000; Scheiring et al., 2020). These processes generate chronic stress and raise the likelihood of harmful coping behaviors such as substance use.

3. Institutional Buffering The mortality consequences of deregulation depend on the institutional context in which it occurs. Health-system capacity, income-support generosity, and exposure to harmful environments shape how economic and psychosocial shocks translate into mortality (Bjorklund, 2023); we refer to this moderating role as institutional buffering and focus on it directly in the empirical design below.

3 Data

This section describes the data sources, variable construction, and sample used in the empirical analysis. Cross-country information on employment protection legislation, mortality outcomes, and institutional characteristics is combined for a panel of 26 OECD economies over 1998–2019. The 26 OECD countries are Australia, Austria, Belgium, Canada, the Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Japan, Luxembourg, the Netherlands, New Zealand, Norway, Portugal, the Republic of Korea, the Slovak Republic, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

3.1 OECD Indicators of Employment Protection

Employment protection is measured using the OECD Employment Protection Indicators, which evaluate the regulations on the dismissal of workers on regular and temporary contracts and the hiring of workers on temporary contracts on a scale from 0 (least strict) to 6 (most strict). Following the conceptual

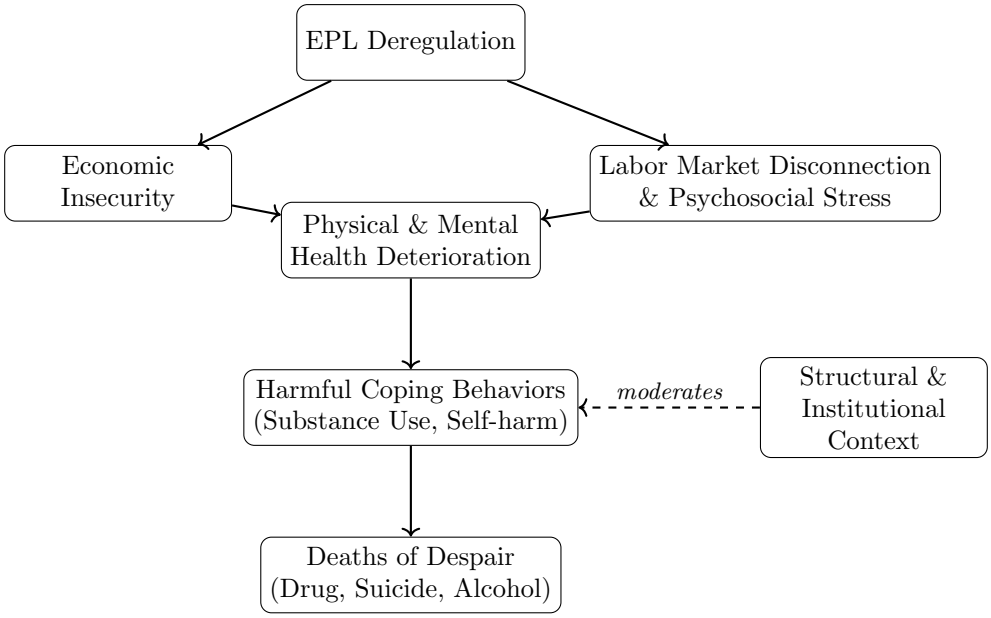


Figure 1: **Conceptual pathways linking employment protection deregulation to deaths of despair.** Solid arrows denote the hypothesized causal channels through which deregulation raises despair-related mortality; the dashed arrow denotes the moderating role of structural and institutional context, which shapes how the health consequences of economic insecurity and psychosocial stress translate into fatal outcomes, rather than itself constituting a direct causal channel.

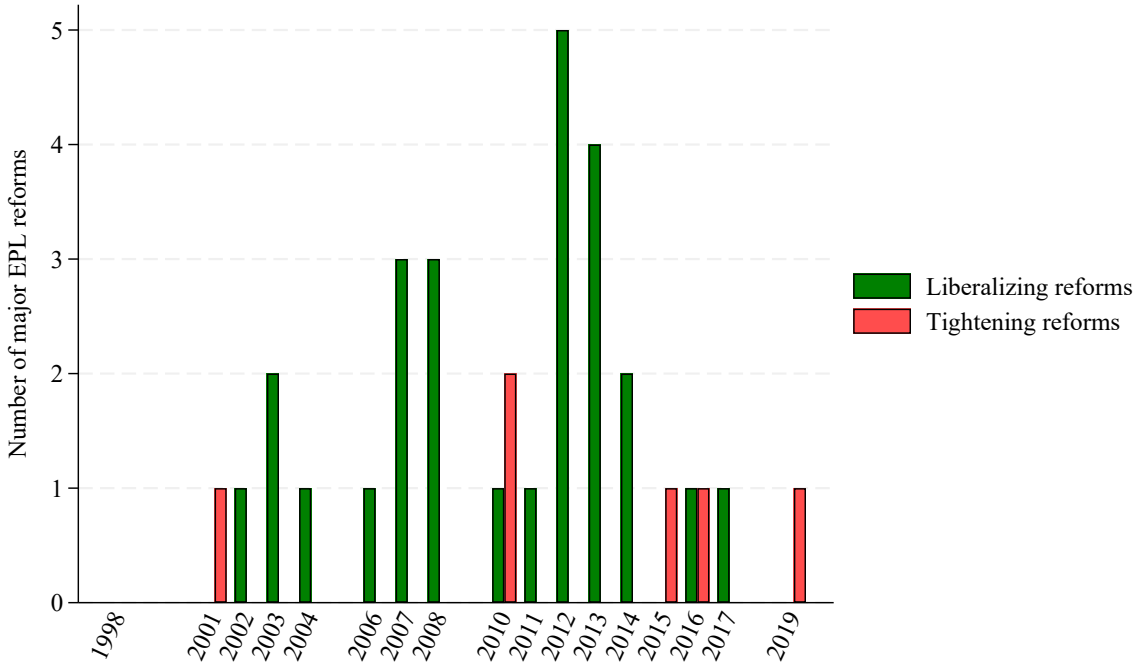
framework, we focus on the dimension of job security, specifically regulations on individual and collective dismissals.

The individual dismissals component of the OECD indicators reflects procedural requirements such as advance notice periods, financial costs such as severance pay obligations, and legal enforceability such as employees' access to legal recourse. The collective dismissals component reflects thresholds for triggering collective dismissal rules, consultation obligations, notification requirements to public authorities, and additional procedural delays.

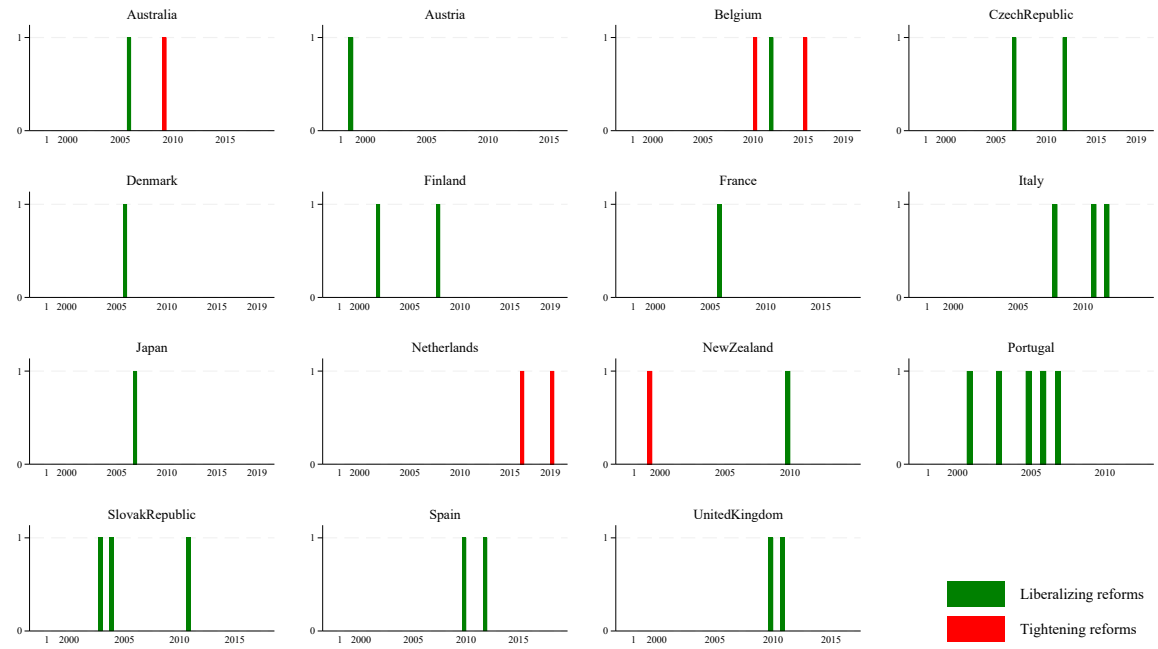
To ensure compatibility with mortality data under tenth revision of the International Classification of Diseases (ICD-10), version 2 of the OECD indicators was used, which provides annual data from 1998 to 2019. Large discrete changes in these indicators were utilized to identify major policy reforms, excluding minor changes in this EPL indicator. Following Ciminelli et al. (2022), major EPL reforms are defined as changes in the indicator that meet the criterion of the top or bottom 5th percentile of the distribution of annual changes. An indicator variable of major EPL reforms is constructed to take the value 1 for major deregulation (liberalizing) reforms, -1 for major tightening reforms, and 0 for no major reforms. This approach isolates substantial policy shifts that can trigger more noticeable mortality responses.

Figure 2 displays the timing and distribution of major EPL reforms across the sample period and countries. The top panel shows the aggregate number of liberalizing and tightening reforms by year. Reforms are unevenly distributed over time, with clustering in the mid-2000s and early 2010s. In particular, liberalizing reforms (deregulation) are most frequent between 2006 and 2013, with a conspicuous peak around 2012, while tightening reforms are relatively infrequent and occur sporadically throughout the sample period.

The bottom panel depicts the distribution of reforms by country, revealing notable cross-country heterogeneity. Some countries, such as the Czech Republic, Finland, Italy, Portugal, the Slovak Republic, Spain, and the United Kingdom, experience multiple liberalizing reforms, whereas others exhibit few or no major reforms over the sample and are dropped from the bottom panel. Tightening reforms are



(a) Distribution of reforms over time



Countries with no major EPL reforms during 1998–2019: Canada, Germany, Greece, Iceland, Ireland, Korea, Luxembourg, Norway, Sweden, Switzerland, UnitedStates.

(b) Reforms by country

Figure 2: **Major employment protection reforms.** The top panel shows the aggregate distribution of reforms over time, while the bottom panel presents the corresponding country-level patterns.

less common and limited to a few countries, including Australia, Belgium, the Netherlands, and New Zealand. The variation in the timing and the number of reforms across countries provides identifying variation for estimating the dynamic effects of EPL deregulation.

3.2 WHO Mortality Database

Mortality data are drawn from the World Health Organization (WHO) Mortality Database, providing age-standardized death rates per 100,000 population. Age standardization ensures comparability across countries with different demographic structures.

Mortality causes associated with deaths of despair are outcome variables, using ICD-10 classifications. The main outcomes are deaths due to drug use disorders, alcohol use disorders, and self-inflicted injuries (suicide). Violence and inflammatory heart diseases are also examined based on a plausible link to substance use and prolonged stress. Cirrhosis of the liver is considered as a supplementary alcohol-related outcome.[1]

To enhance consistency and avoid measurement discontinuities, we restrict the sample to ICD-10 data. Many countries experienced substantial jumps in mortality series during the transition from ICD-9 to ICD-10 because of different classification rules, making ICD-9 and ICD-10 data less comparable. Since most OECD countries adopted ICD-10 around 1998, this restriction yields a balanced and reliable panel covering 26 countries over 1998–2019. Figure 11 in the appendix shows the number of countries by ICD code and year.

3.3 OECD Country-specific Characteristics

To examine heterogeneous effects of major EPL reforms in different institutional settings, country-specific characteristics capturing key institutional and structural conditions are incorporated as moderators. These variables enter the empirical specification as interaction terms with the EPL reform indicator and are normalized to represent differences between the 25th and 75th percentiles.

We consider three broad categories of moderators. First, unemployment benefit replacement rates and government social transfer expenditures are used to represent the extent of income support and economic insurance. Second, avoidable mortality rates and public health insurance coverage are selected to reflect the effectiveness and accessibility of healthcare systems. Third, broader social conditions are proxied by poverty rates, which capture structural disadvantage.

Avoidable mortality, including both preventable and treatable deaths, can represent the effectiveness of public health and healthcare systems. Preventable deaths are defined as causes of death among people aged under 75 years that can be avoided through effective public health and prevention interventions. Treatable deaths are defined as causes of death that can be avoided through timely and effective healthcare interventions, including secondary prevention and treatment (OECD, 2023).

Overall, these moderators allow the current study to assess whether stronger social protection and health institutions mitigate the impact of major EPL reforms on deaths of despair, consistent with the institutional buffering mechanism outlined in Section 2.2.

4 Empirical Methodology

This section specifies the empirical strategy used to estimate the dynamic effects of major employment protection legislation (EPL) reforms on deaths of despair. To capture the timing and persistence of these

[1] ICD-10 codes: drug use disorders (F11–F16, F18–F19, X41–X42, X44); alcohol use disorders (F10, X45); self-inflicted injuries (X60–X84, Y870); violence (X85–Y09, Y871); inflammatory heart diseases (I30–I33, I38, I40, I42); cirrhosis of the liver (K70, K74). Detailed descriptions of these ICD-10 codes are in the appendix.

effects while avoiding restrictive functional form assumptions, the local projection method introduced by Jordà (2005) is employed.

4.1 Baseline Specification

For each horizon $k = 0, \ldots, 5$, we estimate the following specification:

$$y_{i,t+k} - y_{i,t-1} = \alpha_i + \tau_t + \beta_k R_{i,t} + \theta \mathbf{X}_{i,t} + \sum_{h=1}^{k} \phi_h R_{i,t+h} + \sum_{l=1}^{4} (\delta_l \Delta y_{i,t-l} + \gamma_l R_{i,t-l}) + \varepsilon_{i,t}. \quad (1)$$

The dependent variable, $y_{i,t+k}$, denotes the natural logarithm of the age-standardized mortality rate for a given cause of death in country i at horizon k. Taking the logs of the outcome variable makes its distribution closer to a normal distribution. In addition, expressing the outcome in logs lets the coefficients β_k be interpreted as semi-elasticities to approximate percentage changes in mortality. The specification models cumulative changes in log mortality relative to the pre-reform period, allowing the coefficients β_k to be interpreted as impulse responses to EPL reforms according to post-reform horizon k.

The variable $R_{i,t}$ is an indicator of major EPL reform, capturing substantial policy changes identified from the OECD Employment Protection Indicators. Country fixed effects (α_i) control for time-invariant heterogeneity across countries, while year fixed effects (τ_t) capture global shocks affecting all countries simultaneously. The vector $\mathbf{X}_{i,t}$ includes relevant macroeconomic controls such as an economic-crisis dummy, lagged unemployment, union density, and manufacturing share of GDP.

To isolate the effect of reforms occurring at time t, the specification includes forward reform terms $\sum_{h=1}^{k} \phi_h R_{i,t+h}$, which account for additional reforms implemented within the response horizon. Four lags of both the outcome variable and the reform indicator are included to mitigate concerns about reverse causality and dynamic adjustment.

The coefficients β_k trace the dynamic response of mortality to a major EPL reform over time. By estimating the model separately for each horizon k, impulse response functions (IRFs) are generated to describe how mortality responds in the years following a reform. Standard errors are clustered to account for serial correlation.

The identifying assumption is that, conditional on country and year fixed effects and the lagged dynamics built into equation (1), the timing of major EPL reforms is not correlated with the mortality trajectory that would have prevailed absent the reform. Two distinct threats to this assumption are the following. The first is reverse causality running from mortality to policy: reforms could be possibly arranged in response to a mortality crisis. However, major EPL reforms are more likely to be the outcome of political bargaining over labor-market flexibility, typically induced by unemployment, competitiveness, or fiscal considerations, rather than being designed or timed around cause-specific mortality trends. The four lags of the outcome variable are incorporated to address possible reverse causality, and the four lags of the reform indicator and the forward-reform controls in equation (1) further guard against any residual short-run feedback. The second and more subtle threat is confounding by omitted variable bias: even reforms that are not caused by mortality trends may coincide with broader economic or political shifts, such as fiscal consolidation, structural adjustment programs, financial crises, or changes in government, that endogenously impact deaths of despair through channels unrelated to job security. While country and year fixed effects absorb time-invariant country characteristics and common global shocks, but they cannot by themselves rule out that a country undertaking EPL reforms is at the same time undergoing other changes correlated with the mortality response we attribute to deregulation.

We take two complementary steps to assess these threats empirically rather than assume them away.

First, we extend the baseline specification into an event-study design that adds six years of pre-reform leads to the six post-reform horizons, using Teulings–Zubanov-style controls for interim reforms in both directions (Figure 4). If major EPL reforms were confounded with a broader package of other reforms or a country-specific shock that also drives mortality, we would generally expect that confounding factor to leave a footprint like a differential trend on mortality in the years just before or around the EPL reform. Finding instead that pre-reform coefficients are close to zero and show no systematic trend can give more credibility to a genuine mortality response from the EPL reform itself than to a pre-existing or contemporaneously confounded trend. However, it still cannot rule out a confounding shock whose effects happen to coincide exactly with the EPL reform.

Second, the local projections difference-in-differences (LP-DiD) estimator of Dube et al. (2025) is applied to the subsample of 22 countries, excluding countries with major tightening EPL reforms (Figure 7). Unlike the two-way fixed-effects specification in equation (1), LP-DiD estimates each horizon-specific effect using only "clean" comparisons: newly deregulating countries against countries that are not-yet-treated or, under an effect-stabilization assumption, previously treated countries that have re-entered the control pool once their own effects have stabilized. This approach avoids the negative-weighting bias that can affect two-way fixed-effects estimates under staggered treatment timing. This is a useful check on the confounding concern above because it is a different estimator built on a more restrictive set of country-year comparisons than the baseline specification. A result that fails to survive this alternative construction of the control group would raise doubts about whether the baseline estimate reflects a general feature of deregulation episodes or an artifact of a specific comparison. Figure 7 reports five variants of the estimator that differ in how the control sample is constructed: 4a and 4b restrict controls to not-yet-treated countries, with 4a fixing the estimation sample across all horizons to rule out compositional changes in which countries are compared at each horizon, and 4b allowing that sample to vary horizon by horizon; 5a and 5b instead allow previously treated countries to re-enter the control pool once their effects have stabilized, again with (5a) and without (5b) the fixed-sample restriction; and 5c reweights the 5b sample using a regression-adjustment estimator to recover an equally weighted, rather than variance-weighted, average treatment effect. A consistent pattern across all five variants would indicate that the baseline drug-mortality result is not an artifact of the two-way fixed-effects estimator or of any single way of constructing the control group.

4.2 Heterogeneous Effects

To examine how institutional and structural conditions shape the impact of EPL reforms, we extend the baseline specification to include interaction terms:

$$y_{i,t+k} - y_{i,t-1} = \alpha_i + \tau_t + \beta_{1k}(R_{i,t} \times S_{i,t}) + \beta_{2k} R_{i,t} + \beta_{3k} S_{i,t} + \cdots + \varepsilon_{i,t}. \tag{2}$$

Here, $S_{i,t}$ represents country-specific characteristics, including measures of income support, health system capacity, and social conditions. These variables are normalized to reflect differences between countries at the 25th and 75th percentiles. For social transfer expenditure, public health insurance coverage, and the unemployment benefit replacement rate, each of which is naturally increasing in institutional generosity, $S_{i,t}$ is defined as the inverse of the underlying measure so that, like the avoidable death rate and the poverty rate, moving from the 25th to the 75th percentile of $S_{i,t}$ corresponds to moving from a more to a less protective institutional environment for every moderator considered.

The interaction coefficient β_{1k} captures how the effect of major EPL reforms varies across institutional environments. Specifically, it can be interpreted as the differential impact of a major EPL reform between countries at the 25th percentile of $S_{i,t}$ (a more protective environment) and countries at the 75th percentile (a less protective environment). Under this convention, a positive β_{1k} indicates that

the mortality response to deregulation is larger where institutions are less protective, aligned with an institutional-buffering interpretation.

5 Results

5.1 Descriptive Statistics

Table 1 reports summary statistics for the main outcome, treatment, control and moderator, and candidate mediator variables used in the analysis.

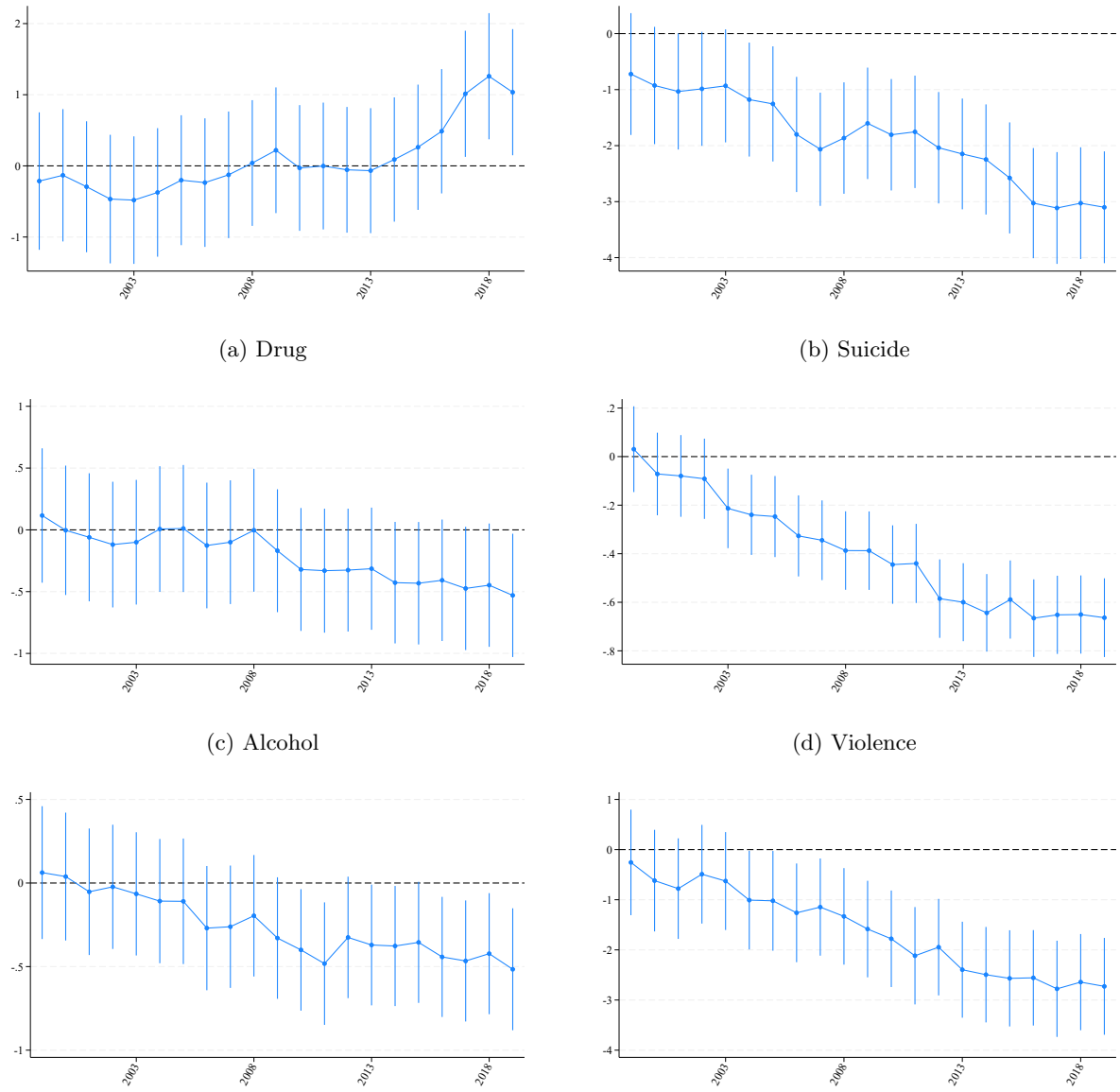
Table 1: Descriptive statistics

Variable	Count	Mean	SD	Min	Max
Outcome variables					
Drug death rate	486	2.561944	2.753695	0.000000	19.09016
Suicide death rate	484	10.895560	4.007673	3.321734	26.31062
Alcohol death rate	486	2.456993	2.144824	0.000000	10.89471
Violence death rate	486	1.140382	1.161581	0.000000	7.333786
Inflammatory heart diseases death rate	488	3.654734	1.323953	1.039139	8.607623
Cirrhosis of liver death rate	488	7.605482	4.283892	1.394737	22.82417
Treatment variable					
EPL reforms	486	0.041152	0.253540	−1	1
Control variables & moderators					
Avoidable death rate	458	220.906100	65.475420	124.000000	515.0000
Replacement rate of UB (60 months)	440	32.854550	16.366160	0.000000	69.0000
Gov. social transfers expenditure (% of GDP)	457	11.893550	3.417891	5.023348	19.67462
Nominal GDP per capita ($)	486	40205.440000	15698.440000	10665.070000	121073.8000
Manufacturing share of GDP	486	15.050760	5.528970	4.554389	34.89132
Unemployment rate	463	7.262532	4.024560	2.013950	26.49027
Poverty rate	267	10.398510	3.362702	4.545772	18.62064
Gov. health insurance coverage rate	462	95.635070	15.335840	22.000000	100.0000
Possible mediators					
Net family earnings, min. wage (% of reference type)	292	55.628770	12.880510	24.640000	80.34000
Disposable income below poverty line (% of population)	257	10.406180	3.354735	4.545772	18.50000
Involuntary part-time employment (% of employment)	458	3.905338	2.318894	0.176000	12.33600
Employment tenure of 1–5 months (%)	393	5.726652	1.808592	2.389000	11.78989

Notes: Mortality rates are per 100,000 population. EPL reforms is the signed reform indicator (−1 for major tightening, 1 for major deregulation, 0 otherwise). The unemployment rate appears both as a control variable and, in horizon-matched cumulative-change form, as a candidate mediator (Section 5.6); it is reported once here. Summary statistics are computed over available observations; counts reflect non-missing country-year observations for each variable, 1998–2019.

Figure 3 summarizes stylized facts on cause-specific mortality across OECD countries. It reports average changes in mortality rates across 26 OECD countries by year, obtained from a specification with country and year fixed effects. Several patterns emerge. Only drug use disorder mortality exhibits an overall upward trend over the sample period, with pronounced increases especially after the mid-2000s. In contrast, alcohol use disorder and inflammatory heart disease mortality display a gradual decline over time, while suicide, violence mortality, and cirrhosis of liver mortality show statistically significant and persistent downward trends. These aggregate patterns highlight heterogeneity across causes of death, validating their separate analysis in the empirical framework.

Appendix Figure 12 shows that these aggregate patterns mask cross-country heterogeneity in the magnitude and direction of mortality trends for each cause of death. Therefore, both cross-country and time-series variation among OECD countries are used to identify the effects of major EPL reforms.



(a) Drug

(b) Suicide

(c) Alcohol

(d) Violence

(e) Inflammatory Heart Disease

(f) Cirrhosis of Liver

Figure 3: **Average changes in cause-specific mortality rates across 26 OECD countries.** Each panel reports the coefficients on the year fixed effects from the regression $Mortality_{i,t} = \alpha + \tau_t + \gamma_i + \varepsilon_{i,t}$, where i and t index country and year, respectively. $Mortality$ denotes the cause-specific mortality rates, α is a constant, τ_t are year fixed effects, and γ_i are country fixed effects. Vertical lines indicate 95% confidence intervals. The estimates can be interpreted as the average change in mortality rates relative to 1998, the omitted base year.

5.2 Main Effects

Figures 4, 5, and 6 report the dynamic effects of major EPL reforms on cause-specific mortality, estimated using the local-projection event-study design of equation (1) extended with six years of pre-reform leads (Section 4.1). Each figure plots both the pre-reform leads and the post-reform response, so the pre-trend evidence and the treatment effect can be read directly off the same plot.

Two main findings are the following. First, as Figure 4 shows, EPL reforms are associated with a statistically and economically significant increase in mortality from drug use disorders. Pre-reform coefficients are flat and close to zero, with no systematic upward or downward trend, while the post-reform response is not immediate: it builds gradually over the post-reform window and becomes both larger and more precisely estimated at longer horizons. By the five-year horizon, drug-related mortality is approximately 35 percent higher than its pre-reform level, indicating a significant medium-run effect rather than a transient shock. This combination of a flat pre-trend and a gradually building post-reform response invalidates reverse causality or pre-existing divergence between reforming and non-reforming countries, and instead supports deeming the post-reform estimates as an impact of EPL reforms rather than an artifact of confounding.

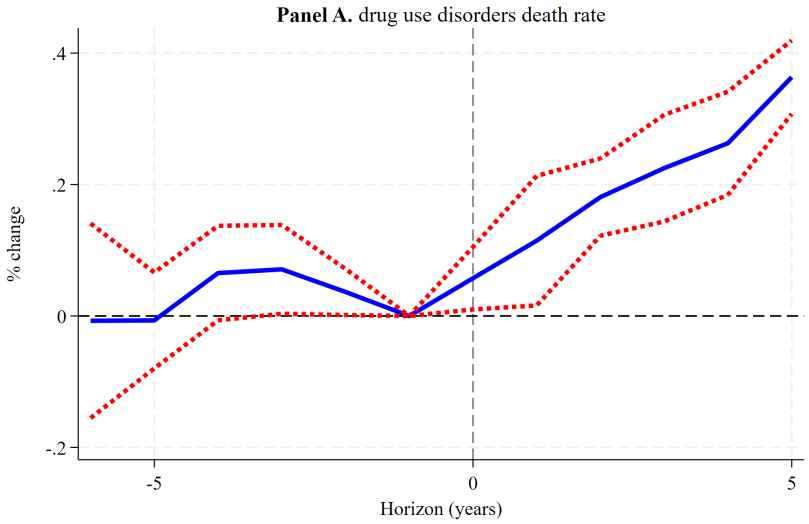


Figure 4: **Dynamic effects of major EPL deregulation on drug use disorder mortality with pre-trend.** The figure plots local-projection event-study estimates of the log mortality rate (per 100,000 population) response to major EPL deregulation for drug use disorder, including six years of pre-reform leads. The horizontal axis reports years relative to reform (horizons $h = -6, \ldots, 0, \ldots, 5$); the reference period is $h = -1$. The solid line shows point estimates; dotted lines show 95% confidence interval bounds. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, *[1998-2019]*.

Second, suicide mortality also rises following major EPL reforms, with similarly flat pre-reform coefficients (Figure 5), although the magnitude is an order of magnitude smaller, reaching roughly 5 percent at the five-year horizon. By contrast, mortality from alcohol use disorder shows a statistically insignificant response at any horizon, before or after reform (Figure 6).

These results point to meaningful heterogeneity across components of deaths of despair: labor market reforms turn out to amplify causes of death with acute substance use and self-harm more than alcohol use disorder. We treat the null result in alcohol use disorder mortality as informative rather than as a meaningless result that should be ignored, and return to it in Section 5.4 with a supplementary analysis of mortality from cirrhosis of liver that is attributed to alcohol.

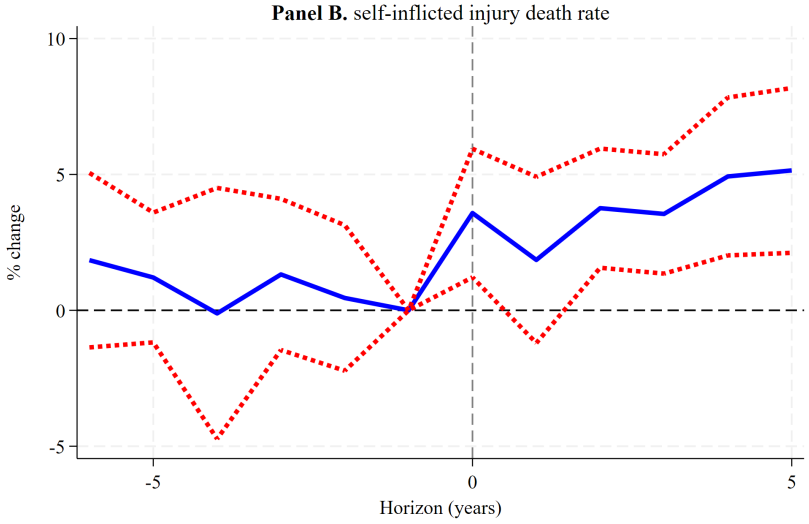


Figure 5: **Dynamic effects of major EPL deregulation on self-inflicted injury (suicide) mortality with pre-trend.** The figure plots local-projection event-study estimates of the log mortality rate (per 100,000 population) response to major EPL deregulation for self-inflicted injury (suicide), including six years of pre-reform leads. The horizontal axis reports years relative to reform (horizons $h = -6, \ldots, 0, \ldots, 5$); the reference period is $h = -1$. The solid line shows point estimates; dotted lines show 95% confidence interval bounds. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, *[1998-2019]*.

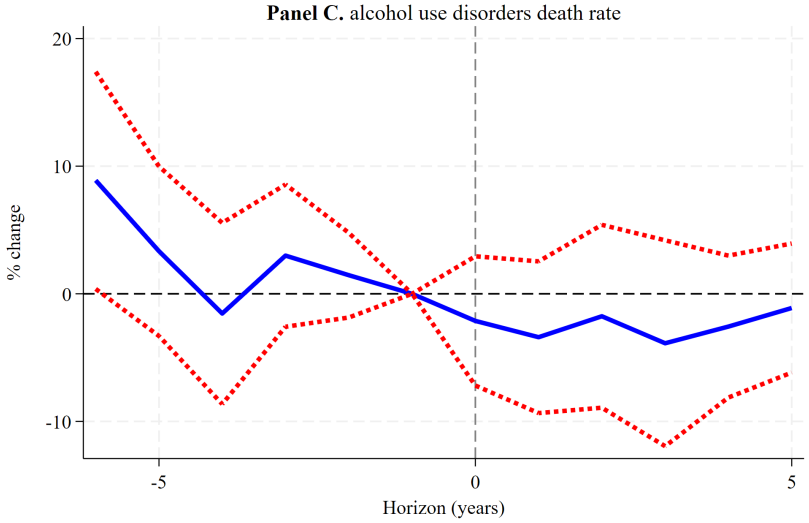


Figure 6: **Dynamic effects of major EPL deregulation on alcohol use disorder mortality with pre-trend.** The figure plots local-projection event-study estimates of the log mortality rate (per 100,000 population) response to major EPL deregulation for alcohol use disorder, including six years of pre-reform leads. The horizontal axis reports years relative to reform (horizons $h = -6, \ldots, 0, \ldots, 5$); the reference period is $h = -1$. The solid line shows point estimates; dotted lines show 95% confidence interval bounds. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, *[1998-2019]*.

5.3 Robustness Checks

Figure 7 reports robustness checks based on the local projection difference-in-differences (LP-DiD) approach of Dube et al. (2025), using only countries experiencing liberalizing reforms. In essence, LP-DiD re-estimates each horizon-specific effect using only "clean" comparisons: newly deregulating countries against countries that are not-yet-treated or that have re-entered the control pool once their own effects have stabilized, rather than the full two-way fixed-effects sample used in the baseline specification. This addresses potential concerns related to treatment heterogeneity, staggered adoption, and contamination from tightening reforms. Furthermore, this approach avoids the negative-weighting bias that can affect two-way fixed-effects estimates when treatment timing is staggered.

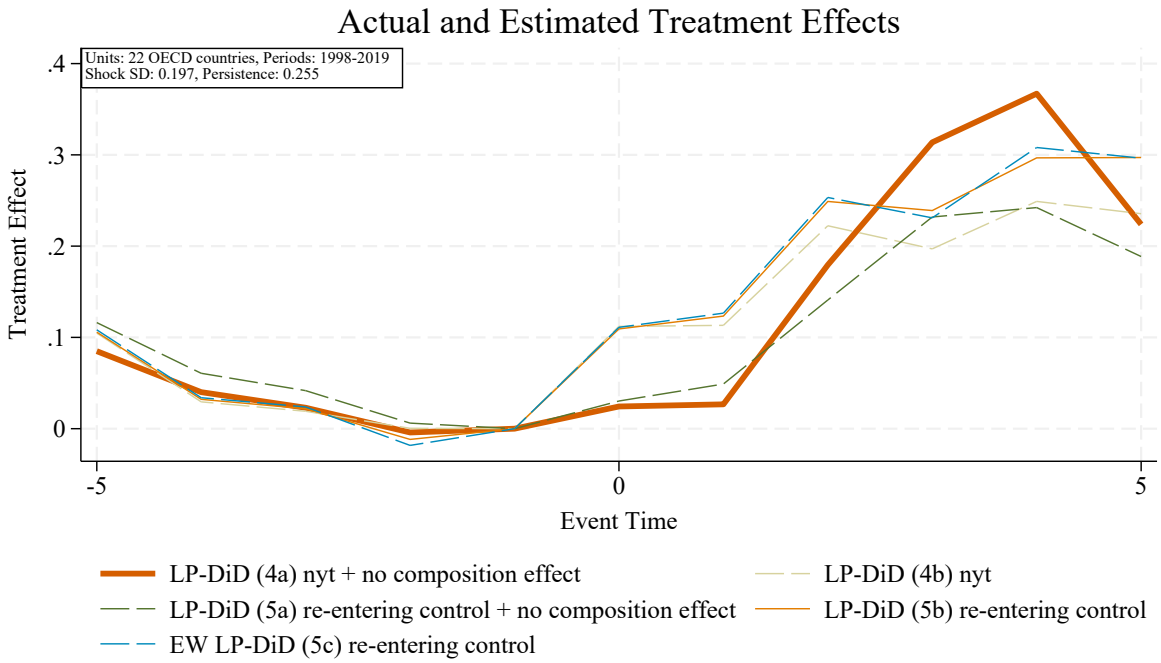


Figure 7: **Robustness check of the effects of major EPL deregulation on drug use disorder mortality with LP-DiD method.** This figure shows LP-DiD estimates of both pre-trend and post-reform response of the log mortality rate (per 100,000 population) for drug use disorder. The horizontal axis reports years before and after reform (horizons $h = -5, \ldots, 0, \ldots, 5$). Solid lines show point estimates; shaded bands are 95% confidence intervals. Standard errors are Driscoll–Kraay. Sample: 22 OECD countries, *[1998-2019]*.

Across four specifications of LP-DiD, the results show a uniform dynamic pattern. Pre-reform coefficients remain close to zero, providing additional support for the absence of differential pre-trends. In contrast, post-treatment effects increase steadily, reaching magnitudes analogous to those obtained in the baseline specification.

It is important to note that the results are robust across different types of LP-DiD estimator. Although some variation in magnitude arises at longer horizons, the qualitative pattern of increasing post-reform effects remains the same. This corroborates that the baseline results are not driven by specification choice or sample composition.

As a further check, Table 2 re-estimates the baseline drug mortality specification adding, one at a time, controls for economic crises, lagged unemployment, the unemployment benefit replacement rate, trade union density, the manufacturing share of GDP, and public social transfer expenditure. The coefficient of interest does not produce any significant discrepancy: point estimates at each horizon move by at

most a few hundredths of a log point relative to the baseline row, and the feature of increasing statistical significance from $h = 0$ through $h = 5$ is preserved throughout. This robust result in mortality from drug use disorder addresses the concern over omitted-variable bias by any one of these macroeconomic or institutional controls.

Table 2: Dynamic effects of major EPL deregulation on mortality from drug use disorders

Controls for	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Baseline	0.06**	0.11**	0.18***	0.22***	0.26***	0.36***
	(0.0224)	(0.0462)	(0.0274)	(0.0375)	(0.0360)	(0.0254)
Crisis dummy	0.05**	0.11**	0.19***	0.22***	0.25***	0.37***
	(0.0237)	(0.0412)	(0.0276)	(0.0330)	(0.0455)	(0.0283)
Lagged unemployment	0.05**	0.12**	0.19***	0.24***	0.27***	0.37***
	(0.0209)	(0.0480)	(0.0257)	(0.0394)	(0.0383)	(0.0312)
Replacement rate of UB	0.06**	0.11**	0.18***	0.22***	0.26***	0.36***
	(0.0223)	(0.0462)	(0.0272)	(0.0370)	(0.0357)	(0.0254)
Union density	0.05***	0.11**	0.18***	0.23***	0.24***	0.34***
	(0.0164)	(0.0474)	(0.0271)	(0.0350)	(0.0545)	(0.0431)
Manufacturing share of GDP	0.06**	0.11**	0.17***	0.20***	0.23***	0.30***
	(0.0217)	(0.0470)	(0.0251)	(0.0369)	(0.0321)	(0.0313)
Social transfers expenditure	0.06**	0.11**	0.18***	0.22***	0.27***	0.38***
	(0.0218)	(0.0443)	(0.0261)	(0.0343)	(0.0422)	(0.0445)

Notes: Outcome is the log mortality rate from drug use disorders (per 100,000 population). Entries report dynamic semi-elasticities at horizons $h = 0$ to 5 years after major EPL deregulation; standard errors in parentheses. Stars denote significance: *$p < 0.10$, *$p < 0.05$, *$p < 0.01$.

5.4 Effects on Other Causes of Death

Effects on other causes of death such as violence, inflammatory heart diseases, and cirrhosis of the liver are examined next. Mortality from violence and inflammatory heart diseases are examined based on the linkage to substance use and accumulated stress, and mortality caused by cirrhosis of the liver is explored on the background of the null effect of major EPL reforms on alcohol use disorder mortality. Figures 8 (Panel D), 9 (Panel E), and 10 (Panel F) present local-projection event-study estimates, including six years of pre-reform leads, for three additional outcomes: cirrhosis of the liver, violence, and inflammatory heart disease.

Figure 8 shows that cirrhosis mortality is statistically indistinguishable from zero at nearly every horizon, both before and after reform. A single point estimate (at $h = 4$, approximately 3 percent) reaches significance, but it is not part of a systematic pattern: it does not build steadily across horizons the way the drug and suicide effects do, and it fades by $h = 5$. This is interpreted as a null result for cirrhosis of the liver, parallel with the null result for alcohol use disorder mortality documented in Section 5.2.

The joint null result for alcohol use disorder and cirrhosis mortality is viewed as a result that guides the interpretation of other estimates. Unlike drug and suicide mortality, which rise steadily and significantly through year 5, neither outcome shows a cumulative and persistent response to deregulation. This indicates that the mortality response to major EPL reforms is concentrated in and most precisely estimated for causes of death with comparatively rapid behavioral and physiological onset (drug overdose and suicide) compared to the alcohol-related pathway where 10 to 20 years are typically required for alcohol-related liver disease to develop. The five-year post-reform window may be too short to detect an effect even if one exists.

Turning to violence and inflammatory heart disease, Figures 9 and 10 show that EPL deregulation is

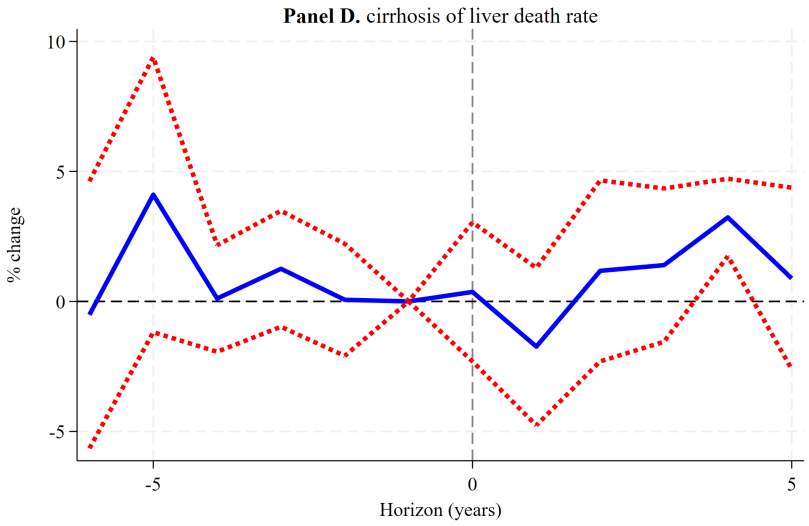


Figure 8: **Dynamic effects of major EPL deregulation on cirrhosis of the liver mortality with pre-trend.** The figure plots local-projection event-study estimates of the log mortality rate (per 100,000 population) response to major EPL deregulation for cirrhosis of the liver, including six years of pre-reform leads. The horizontal axis reports years relative to reform (horizons $h = -6, \ldots, 0, \ldots, 5$); the reference period is $h = -1$. The solid line shows point estimates; dotted lines show 95% confidence interval bounds. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, *[1998-2019]*.

associated with increases in mortality from both causes that become statistically significant only at the four- and five-year horizons, reflecting the delayed pattern documented for drug mortality. Increases in substance use and prolonged stress may contribute to both heightened exposure to violent environments and elevated cardiovascular risk.

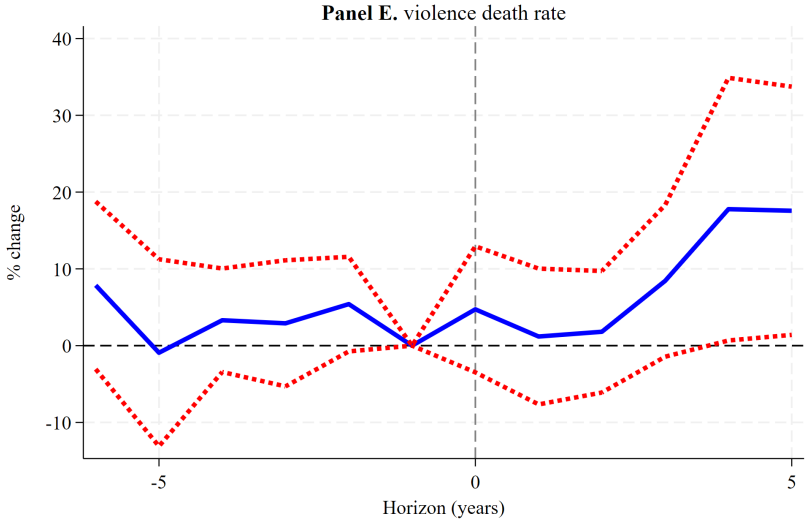


Figure 9: **Dynamic effects of major EPL deregulation on violence mortality with pre-trend.** The figure plots local-projection event-study estimates of the log mortality rate (per 100,000 population) response to major EPL deregulation for violence, including six years of pre-reform leads. The horizontal axis reports years relative to reform (horizons $h = -6, \ldots, 0, \ldots, 5$); the reference period is $h = -1$. The solid line shows point estimates; dotted lines show 95% confidence interval bounds. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, *[1998-2019]*.

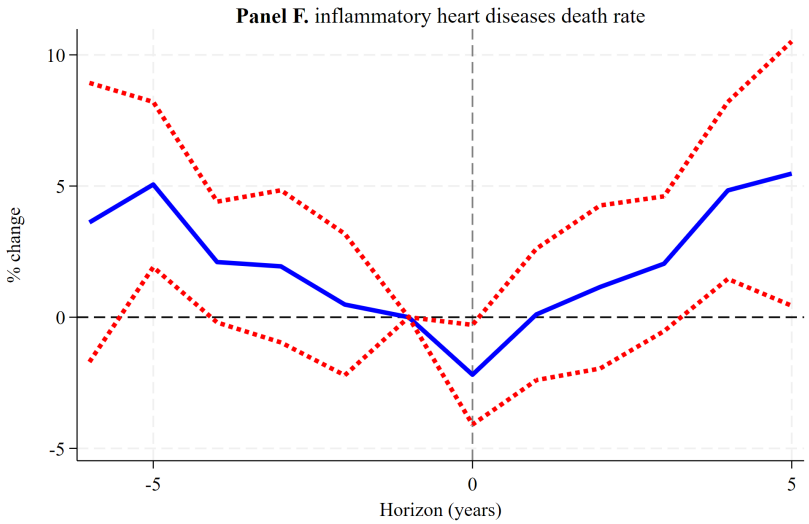


Figure 10: **Dynamic effects of major EPL deregulation on inflammatory heart disease mortality with pre-trend.** The figure plots local-projection event-study estimates of the log mortality rate (per 100,000 population) response to major EPL deregulation for inflammatory heart disease, including six years of pre-reform leads. The horizontal axis reports years relative to reform (horizons $h = -6, \ldots, 0, \ldots, 5$); the reference period is $h = -1$. The solid line shows point estimates; dotted lines show 95% confidence interval bounds. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, *[1998-2019]*.

5.5 Heterogeneous Effects by Institutional Context

The average responses reported in Sections 5.2 and 5.4 pool countries with fairly different social-insurance and health-system structures. To test whether the mortality response to major EPL reforms depends on this institutional context, the baseline specification needs to be extended with an interaction between the reform indicator and each of five country characteristics: the avoidable death rate, social transfer expenditure, public health insurance coverage, the unemployment benefit replacement rate by 60 months, and the poverty rate (equation (2), Section 4.2). Each moderator is normalized so that the interaction coefficient β_{1k} is interpretable as the differential mortality response between a country at the 75th percentile of the moderator and one at the 25th percentile. Tables 3, 6, 7, 8, and 9 report the resulting coefficients at each horizon for drug use disorders, suicide, alcohol use disorder, violence, and inflammatory heart disease mortality, respectively.

For drug mortality, the cause of death with the largest and most precisely estimated average effect, institutional context matters along several dimensions (Table 3). The avoidable death rate interaction is positive and significant from the two-year horizon onward, rising from 0.34 to 0.50 log points ($p < 0.05$); the public health insurance coverage interaction is positive and significant at every horizon, growing from 0.01 to 0.17 log points; and the poverty rate interaction is negative and significant from impact through the four-year horizon, reaching -0.62 log points at $h = 3$ before losing significance at $h = 5$. Social transfer expenditure and the unemployment benefit replacement rate do not interact significantly with major EPL reforms at any horizon for this outcome.

For the remaining four outcomes, full results are reported in Appendix Tables 6–9. Suicide interactions are estimated far less precisely than those for drug mortality, which can be expected with the smaller average effect documented in Section 5.2. Social transfer expenditure is the only moderator of suicide mortality, which is positive and significant at more than one horizon, implying that less social transfer expenditure by governments leads to higher suicide mortality by about 18%. While alcohol use

disorder mortality shows no significant *average* response to major EPL reforms (Section 5.2), an institutional interaction with social transfers expenditure is significant in its negative sign at four horizons, reaching around a 33% decrease 4 years after a major EPL reform. A less generous replacement rate of unemployment benefit is also associated with increasing alcohol use disorder mortality at the two-year and three-year horizons after a major EPL reform. This suggests that offsetting responses across more and less generous welfare states may mask an average effect for this cause of death. Violence and inflammatory heart disease mortality, the two additional outcomes discussed in Section 5.4, display institutional interactions concentrated at the horizons where their average effects also emerge. The replacement rate of unemployment benefits is a significant moderator of violence mortality with a positive sign, reaching a large increase of around 54% with less generous unemployment benefits. While the avoidable death rate is a significant moderator of inflammatory heart disease mortality with a positive sign, reaching around 8% increase with a higher avoidable death rate in a less effective health care system, a higher poverty rate is associated with a significant decrease in inflammatory heart disease mortality up to around 42%.

Table 3: Dynamic effects of EPL deregulation on mortality from drug use disorders: interaction terms

Interaction term with	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Avoidable death rate	0.14**	0.22	0.34**	0.50**	0.40**	0.40**
	(0.0544)	(0.1279)	(0.1372)	(0.1699)	(0.1785)	(0.1731)
Social transfers expenditure	0.05	0.14	0.05	0.17	0.35	0.49
	(0.0843)	(0.1615)	(0.2287)	(0.2829)	(0.3316)	(0.3240)
Public health insurance coverage	0.01***	0.01*	0.04**	0.08**	0.11*	0.17*
	(0.0036)	(0.0082)	(0.0156)	(0.0324)	(0.0573)	(0.0836)
Replacement rate of UB (60 months)	0.09	0.22	0.02	0.05	-0.06	0.06
	(0.0957)	(0.1887)	(0.2097)	(0.2237)	(0.2350)	(0.2046)
Poverty rate	-0.19*	-0.47**	-0.54***	-0.62**	-0.51**	-0.19
	(0.0947)	(0.1934)	(0.1593)	(0.2051)	(0.1812)	(0.1384)

Notes: Outcome is the **log mortality rate from drug use disorders** (per 100,000 population). Entries are interaction effects (P75–P25 style) at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.6 Candidate Channels

Section 2.2 outlines a conceptual map from major EPL reforms to deaths of despair through economic insecurity, labor market disconnection, and institutional buffering channels. The institutional buffering channel is empirically examined in Section 5.5. To probe the other mechanisms more directly, it is tested whether a set of candidate mediating variables (proxies for material strain and for labor market disconnection) responds to major EPL reforms with a timing pattern comparable to the one estimated for drug mortality. For each candidate M, we estimate

$$M_{i,t+k} - M_{i,t-1} = \alpha_i + \tau_t + \beta_k R_{i,t} + \theta \mathbf{X}_{i,t} + \sum_{h=1}^{k} \phi_h R_{i,t+h} + \sum_{l=1}^{4} (\delta_l \Delta M_{i,t-l} + \gamma_l R_{i,t-l}) + \varepsilon_{i,t}$$

for $k = 0, \ldots, 5$, using the identical horizon-matched cumulative-change transformation, lag structure, and forward-reform correction used in equation (1) for mortality but applied to the candidate mediator. This puts the question "does M respond to major EPL reforms, and on what timeline" on the same footing as the mortality regression itself. Table 4 reports results for five candidates spanning both proposed mechanisms: net family income for a low-wage family and the poverty rate (material strain); the short-tenure employment share, the aggregate unemployment rate, and the involuntary part-time

employment share (labor market disconnection).

Results show that two candidates move with a timing and sign compatible with the conceptual framework. The poverty rate rises following major EPL reforms, with the response building over time and reaching statistical significance only at the four- and five-year horizons (0.82 and 1.43 percentage points, $p < 0.05$ and $p < 0.01$, Table 4), the same horizons at which the drug mortality response in Table 2 is largest and most precisely estimated. The short-tenure employment share (the fraction of employees with one to five months of job tenure) rises significantly from the two-year horizon onward (0.36 to 0.45 percentage points, $p < 0.10$ or better), congruent with the conceptual expectation of increasing job churn and reducing the sense of stable labor market attachment that the psychosocial-disconnection channel in Section 2.2 emphasizes. Net family income for a minimum-wage couple with two children moves in the theoretically expected (negative) direction from the second horizon onward and is directionally consistent throughout. However, the response is noisier and only marginally significant at $h = 2$ and $h = 5$ ($p = 0.066$ and $p = 0.088$), which is likely reflecting the smaller sample of countries for which this measure is available.

The remaining two candidates move in the direction opposite to a naive prediction. Both the aggregate unemployment rate and the involuntary part-time employment share *decline* following major EPL reforms, and increasingly so: the unemployment rate falls by up to 1.16 percentage points by the five-year horizon ($p < 0.05$), and involuntary part-time employment falls by up to 0.91 percentage points ($p < 0.05$). Taken together with the short-tenure result above, the results across all five candidates argue against a crude story in which major EPL reforms raise drug mortality or suicide mortality simply by raising joblessness or involuntary underemployment; if anything, conventional labor-market-slack measures improve after EPL reforms. Instead, the evidence points toward rising material strain (poverty) and rising job instability (short-tenure churn), independent of the aggregate quantity of employment, as the more plausible channels.

These results are reported as suggestive screening evidence, not as a formal causal mediation decomposition. Confirming that a candidate variable responds to major EPL reforms is a necessary but far from sufficient condition for it to be a genuine mediator of the mortality effect: a formal decomposition into natural direct and indirect effects would additionally require sequential ignorability of the mediator conditional on treatment and observed confounders, an assumption that is regarded as not obvious in a 26-country annual panel where mediators and mortality outcomes are likely to share unobserved country-year confounders not captured by our controls. This exercise is viewed as motivating for future mediation analysis using individual- or household-level panel data with richer information on employment transitions, income, and health-seeking behavior.

5.7 Asymmetry Between Deregulation and Tightening

The baseline specification codes major EPL reforms with a single signed indicator (+1 for deregulation, −1 for tightening; Section 3.1), which imposes by construction that tightening reverses the mortality consequences of deregulation exactly: a single coefficient β_k at each horizon applies with the opposite sign to both reform types. This symmetry restriction need not hold. If job protections operate partly through channels with hysteresis, for instance if workers who lose employment during a period of weaker protection do not simply regain it once protections are restored, or if the substance-use and mental-health consequences of a deregulation episode persist independently of subsequent policy reversals, then tightening would be expected to have a smaller, or even a same-signed, effect rather than a mirror-image one.

This possible asymmetry is tested directly by decomposing the signed indicator into two non-negative dummies, one for major deregulation episodes and one for major tightening episodes, and entering both

Table 4: Timing of candidate mediating variables following major EPL deregulation

Candidate mediator	0y	1y	2y	3y	4y	5y
Net family income (minimum-wage couple, 2 kids)	-0.27	-1.41	-2.49*	-3.11	-4.20	-5.27*
	(0.369)	(0.898)	(1.248)	(2.416)	(2.881)	(2.837)
Poverty rate	0.51	0.45	0.62	0.45	0.82**	1.43***
	(0.503)	(0.317)	(0.560)	(0.490)	(0.358)	(0.340)
Short-tenure employment share (1–5 mo.)	-0.01	0.15	0.36**	0.40**	0.45*	0.44*
	(0.087)	(0.148)	(0.158)	(0.168)	(0.227)	(0.228)
Unemployment rate	-0.01	-0.22	-0.50*	-0.66**	-1.01*	-1.16**
	(0.169)	(0.259)	(0.270)	(0.303)	(0.508)	(0.480)
Involuntary part-time employment	-0.01	-0.15	-0.11	-0.48	-0.74*	-0.91**
	(0.123)	(0.165)	(0.278)	(0.286)	(0.378)	(0.356)

Notes: Each row reports $\hat{\beta}_k$ from a separate regression of the candidate mediator's own horizon-matched cumulative change on the EPL deregulation indicator, following the specification described in the text (identical lag structure and forward-reform correction to equation (1), applied to the mediator rather than to mortality). Standard errors, clustered by country, in parentheses. Significance: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

jointly in place of the single signed indicator, with their own lags and horizon-specific forward-term corrections. Table 10 and Figure 13 (Appendix) report the results for drug use disorder mortality. The deregulation coefficients closely track the baseline estimates in Table 2 and are significant at every horizon. The tightening coefficients are negative from year 2 onward, following the symmetric expectation that tightening partially reverses the mortality consequences of deregulation, but they are estimated far less precisely and are statistically significant only at the three- and five-year horizons. This result is based on the fact that tightening episodes are considerably fewer in our sample and concentrated in a handful of countries including Australia, Belgium, the Netherlands, and New Zealand (Section 3.1).

Evidence against exact symmetry is found, but it points in two different directions at different horizons rather than toward a single and uniform pattern. At impact ($h = 0$), the tightening coefficient (0.077) is positive rather than negative (the same sign as the deregulation coefficient, 0.092), and a formal test of H_0: $\beta_{dereg} + \beta_{tighten} = 0$ rejects exact symmetry at the 5 percent level ($p = 0.020$). Seemingly, this is aligned with a short-run one-way ratchet: tightening does not immediately lower drug mortality in the way deregulation immediately raises it. At the five-year horizon, symmetry is rejected again ($p = 0.016$), but in the opposite direction: the tightening coefficient (-0.955) is more than twice the magnitude of the deregulation coefficient (0.396), so tightening is associated with a significantly larger mortality decline than deregulation is with a mortality increase. At the intervening horizons ($h = 1$ through $h = 4$), point estimates are of the theoretically expected negative sign but imprecisely estimated, and symmetry cannot be rejected.

In conclusion, the findings suggest that a single symmetric coefficient understates the complexity of how two different EPL reforms propagate into mortality. Characterizing this asymmetry more precisely than the current study allows is a natural direction for future research, ideally with a larger or more disaggregated set of tightening episodes drawn from a longer sample period or a broader set of countries.

6 Discussion

6.1 Interpretation of Mechanisms

The results speak directly to the three mechanisms outlined in the conceptual framework (Section 2.2): economic insecurity and material strain, labor market disconnection and psychosocial stress, and interaction with structural and institutional context. The dynamic pattern documented in Table 2 is itself informative about which channels are likely operative: the baseline semi-elasticity of drug mortality rises

monotonically from 0.06 log points at impact to 0.36 log points at the five-year horizon, a gradual and cumulative pattern that possibly points to a slow accumulation of economic and psychological strain. This accumulation over time is consistent with the related literature: Jolivet and Postel-Vinay (2025) document that the mental-health consequences of job loss unfold gradually over the life cycle rather than resolving quickly; Scheiring et al. (2020) show that deindustrialization operates as a slow-acting, cumulative driver of population health.

The fact that suicide responds more modestly (roughly 5 percent at five years) than drug mortality (roughly 35 percent) is also suggestive regarding the mechanisms. Both causes presumably share the same underlying economic-insecurity channel, but drug mortality is also sensitive to the local drug-supply environment (the availability, potency, and price of illicit substances), which can amplify the health consequences of a given increase in psychosocial distress; suicide has no comparable supply-side amplifier. The same shared pathway can be derived from the timing of the results for these additional causes of death: violence and inflammatory heart disease mortality rise only at the four- and five-year horizons (Section 5.4), mirroring the delayed drug mortality response.

The clinical and epidemiological literature offers an account of why sustained substance use would raise mortality from inflammatory heart disease and violence with a lag rather than immediately. Chronic heavy alcohol consumption is strongly associated with toxic myocarditis, which is a type of heart inflammation (Carditis) in addition to endocarditis and pericarditis (Piano & Phillips, 2023); stimulants are linked to myocarditis through catecholamine surge and direct toxicity (Virmani et al., 2001); and opioid and intravenous drug use are leading causes of infective endocarditis in young adults (Schmidt et al., 2022). Because these cardiac conditions take time to develop into fatal events, this pathway is consistent with the delayed emergence of the inflammatory heart disease response. Alcohol and certain drugs are also found to elevate the risk of homicide and assault through impaired judgment, aggression, and involvement in illicit drug markets (Darke, 2010; Foran & O'Leary, 2008; Kuhns et al., 2014; World Health Organization, 2018), supporting a reading of the delayed violence and inflammatory heart disease responses as downstream consequences of the same rise in substance use that drives the primary drug mortality result.

Additional evidence to infer mechanisms is from Table 2. By adding control variables one at a time to the baseline specification, change in the magnitude of the impact of major EPL reforms on drug use disorder mortality can be observed. Based on results at the five-year horizon, union density and manufacturing share of GDP reduce the coefficient of major EPL reforms from the baseline level of 0.36 to 0.34 and 0.30, respectively. Other controls keep the baseline coefficient of major EPL reforms maintained or increase it to 0.37 or 0.38. A possible mediation of drug use disorder mortality through manufacturing share of GDP is outstanding among other control variables. It can be also linked to both channels of economic insecurity and labor market disconnection. Declining share of manufacturing sectors in GDP is related to a fewer stable and high-income jobs to workers.

6.2 Alternative Explanations

A natural alternative explanation for the observed rise in drug mortality is a supply-side account, in which changes in the illicit drug environment (the substitution from prescription opioids to heroin and, after roughly 2013, to illicitly manufactured fentanyl) drive the increase (Ruhm, 2022; Ruhm, 2018). Several features of our results comment on this possibility. First, the identification strategy exploits the staggered timing of major EPL reforms across 26 countries (Figure 2), whereas the opioid-to-fentanyl transition is heavily concentrated in North America with limited penetration in Southern and Eastern European labor markets. Furthermore, because our specification estimates the mortality response relative to each country's own pre-reform path, a common supply shock affecting all countries simultaneously cannot

mechanically generate the estimated dynamic response. Second, the event-study evidence in Figure 4 shows flat and statistically insignificant pre-reform coefficients, which would not be obtained if the drug environment happened to deteriorate systematically just before reforms. Third, the age-specific results in Table 11 show effects concentrated among prime-age and older workers (ages 20–69), with no significant increase among the youngest cohort (15–19). This is more congruent with a channel operating through employment and its attendant economic security than with a supply-driven fentanyl narrative that has disproportionately affected younger users in North America.

However, this study cannot entirely rule out that reforms are correlated with unobserved and time-varying local drug-market conditions, particularly where major deregulation episodes and the fentanyl-driven overdose wave coincided in the 2010s. Thus, the overall evidence supports Case and Deaton (2022)'s synthesis in which demand-side distress and supply-side drug availability interact rather than operate as competing explanations: weakening labor market institutions presumably increase the population at risk of problematic substance use while the potency and availability of the local drug supply shape how that latent vulnerability translates into fatal outcomes. This is an interpretation reinforced by the institutional heterogeneity discussed next, which a purely supply-driven account would not predict.

6.3 Institutional Buffering

Building on the results reported in Section 5.5, it is questioned whether institutional context moderates the mortality response to major EPL reforms in a way that admits a coherent causal interpretation. The evidence supports an institutional-buffering interpretation, but it is more nuanced than a single moderator operating uniformly across causes of death. Health-care capacity, represented by avoidable death rate, is a clear buffer for the mortality of drug use disorder and inflammatory heart diseases. Countries with higher avoidable death rates experience significantly larger mortality responses to major EPL reforms for both drug and inflammatory heart disease mortality. Income-support generosity buffers the two outcomes, suicide and violence, as a protective story would predict, but for alcohol use disorder mortality it runs the opposite way, with more generous transfers and unemployment benefits associated with a *larger* response. One possible interpretation is that, unlike suicide and violence, alcohol consumption may be constrained less by psychological distress than by purchasing power: income support need not reduce the propensity to drink under economic stress of EPL reforms so much as it preserves the disposable income that job loss and income instability would otherwise curtail, sustaining consumption that a binding budget constraint would suppress in less generous welfare states. Under this affordability channel, weaker income support would dampen the alcohol-mortality response not through psychological buffering but by limiting the resources available to induce it.

The poverty rate interaction is significantly negative at multiple horizons for drug use disorder mortality and inflammatory heart diseases. It is most pronounced for drug mortality, negative and significant at five of six horizons (Table 3). Added-variable and leave-one-country-out checks confirm this specific coefficient is not driven by any single country (Appendix Figures 14 and 15). This result with the institution of poverty rate can be interpreted with the same affordability channel discussed above for income support and alcohol: poverty constrains the purchasing power needed to turn the psychological and material pressures that major EPL reforms generate into substance use, so a rise in economic insecurity translates less readily into drug use and hence drug mortality, where households already face a binding budget constraint. This logic presumably extends to inflammatory heart disease mortality, whose poverty interactions are also negative and significant at several horizons (Appendix Table 9): to the extent these outcomes are downstream consequences of substance use rather than independent channels (Section 6.1), a binding budget constraint on substance use would be expected to dampen its mortality response too.

The heterogeneity results reveal a layered story: health-system capacity as a buffer for drug use

disorder and inflammatory heart disease mortality; income-support generosity and the poverty rate as two sides of the same affordability channel, with income support buffering the acute-crisis outcomes (suicide, violence) but amplifying alcohol mortality, and poverty dampening drug and inflammatory heart disease mortality by constraining the resources needed to respond to distress with substance use.

6.4 Policy Implications

These findings have policy implications for how labor market reforms are evaluated and designed. Standard cost-benefit assessments of EPL reforms typically weigh efficiency gains (reduced hiring frictions, improved allocative efficiency, lower structural unemployment) against distributional costs measured in wages or the labor share of income (Barbieri & Cutuli, 2016; Ciminelli et al., 2022). The current study suggests that this accounting is incomplete: a five-year, 35 percent increase in drug mortality and a 5 percent increase in suicide mortality following major EPL reforms represent a public-health externality that goes beyond conventional labor market metrics but is arguably comparable in welfare terms to the employment gains such reforms are designed to achieve.

The heterogeneity results offer a constructive policy lesson. Because the adverse mortality response from drug use disorder and inflammatory heart diseases is significantly attenuated in countries with more effective health systems, and because suicide- and violence-related harms are offset in countries with more generous income support, specific and actionable complements to flexibility-enhancing reforms are suggested instead of a blanket argument against labor market deregulation. Countries pursuing EPL reform could accommodate expanded access to addiction treatment and overdose-reversal services, continuity of health coverage for displaced workers, and adequately funded unemployment insurance, particularly given the concentration of estimated effects among prime-age and older workers who are more likely to have dependents and less time to recover from labor market displacement (Table 11). In conclusion, the results support treating employment protection legislation, unemployment insurance, and health-system capacity as interconnected components of a country's social-insurance architecture, echoing calls in the literature to integrate labor market and public-health policy design (Beseran et al., 2022; Kuo & Kawachi, 2023).

6.5 Limitations

Several limitations qualify these conclusions and suggest directions for future research. First, and most fundamentally, the analysis is conducted at the country-year level; it cannot be observed which individuals lose employment protection, whether they experience job loss, or the pathway from labor market events to fatal outcomes. Our estimates therefore capture aggregate population-level responses instead of individual-level treatment effects.

Second, the definition of a "major" reform (changes in the OECD EPL indicator falling in the top or bottom 5th percentile of the annual-change distribution, following Ciminelli et al. (2022)) is a useful but somewhat arbitrary threshold, and it collapses considerable heterogeneity in reform content (e.g., changes to individual versus collective dismissal rules, notice periods versus severance costs) into a single binary indicator. Reforms of similar magnitude on the OECD index may differ quite much in their practical labor market consequences, and the current design cannot discern which specific regulatory margins are most consequential for health.

Third, while the local projection design with country and year fixed effects, lagged outcomes and treatments, and forward reform controls is intended to mitigate reverse causality and confounding from contemporaneous shocks, it is hard to completely rule out unobserved and time-varying country-specific shocks such as political transitions, macroeconomic crises, or shifts in the drug-supply environment that coincide with reform timing in ways not fully captured by our controls. The LP-DiD robustness check

(Figure 7) addresses concerns about staggered-adoption bias and contamination from tightening reforms specifically, but shares this broader limitation.

Fourth, cross-country comparability of cause-specific mortality coding is an important caveat. Restricting the sample to the ICD-10 era (post-1998) improves comparability relative to spanning the ICD-9/ICD-10 transition, but countries continue to differ in coroner and medical examiner practices, and misclassification between suicide, accidental poisoning, and undetermined intent is well documented in the mortality literature; to the extent that such misclassification correlates with reform timing or with our institutional moderators, it could bias our estimates in either direction. In a similar vein, the sample of 26 OECD economies over 1998–2019 excludes the most recent fentanyl-driven surge in North American overdose mortality after 2019 and is not necessarily representative of labor markets outside the OECD; it is not appropriate to extrapolate the estimates to lower- and middle-income countries with different labor market institutions and drug environments.

Fifth, statistical power varies across the heterogeneous-effects specifications: the poverty rate interaction, for example, is estimated on 267 country-year observations compared with up to 486 in the baseline specification (Table 1), reflecting more limited data availability for this moderator. Combined with the fact that the institutional heterogeneity analyses examine five outcomes and five institutional moderators across six horizons, some caution is necessary regarding multiple comparisons, and the moderator results discussed in Section 6.3 should be understood as suggestive of institutional heterogeneity.

7 Conclusion

This paper asks whether labor-market institutions that shape job security and income stability have measurable consequences for despair-related mortality. Using a panel of 26 OECD economies over 1998–2019 and a local-projection design applied to major reforms of the OECD Employment Protection Indicators, it is found that major EPL reforms are followed by a statistically and economically significant, gradually accumulating rise in mortality from drug use disorders and suicide: drug-related deaths increase by approximately 35 percent and suicide mortality by approximately 5 percent within five years of reform, with no detectable average effect on alcohol-related or cirrhosis mortality, and delayed increases in violence and inflammatory heart disease mortality. These headline findings passed a set of robustness checks, including event-study evidence of flat pre-reform trends, an alternative local-projection difference-in-differences estimator, controls for a range of contemporaneous macroeconomic and institutional factors, and a test for asymmetry between deregulation and tightening episodes.

The results are aligned with the conceptual channels of economic insecurity and job instability that run from labor-market institutions to health outcomes, but the country-level design cannot establish more thorough mechanisms and cannot strictly separate labor-market effects from contemporaneous country-specific shocks. Institutional heterogeneity provides suggestive evidence that health-system capacity, income-support policies, and poverty rate may moderate the mortality response from different causes differently.

To conclude, while evaluations of labor-market reforms generally focus on their impacts on employment, productivity, and wages, the current study suggests there may also be health consequences worth incorporating into how such policies are evaluated.

8 References

References

Barbieri, P., & Cutuli, G. (2016). Employment protection legislation, labour market dualism, and inequality in europe. *European Sociological Review*, *32*(4), 501–516. https://doi.org/10.1093/esr/jcv058

Benach, J., Vives, A., Amable, M., Vanroelen, C., Tarafa, G., & Muntaner, C. (2014). Precarious employment: Understanding an emerging social determinant of health. *Annual Review of Public Health*, *35*(1), 229–253. https://doi.org/10.1146/annurev-publhealth-032013-182500

Beseran, E., Pericàs, J. M., Cash-Gibson, L., Ventura-Cots, M., Porter, K. M. P., & Benach, J. (2022). Deaths of despair: A scoping review on the social determinants of drug overdose, alcohol-related liver disease and suicide. *International Journal of Environmental Research and Public Health*, *19*(19), 12395. https://doi.org/10.3390/ijerph191912395

Bjorklund, E. (2023). The needle and the damage done: Deaths of despair, economic precarity, and the white working-class. *Social Science & Medicine*, *333*, 116153. https://doi.org/10.1016/j.socscimed.2023.116153

Bradford, A. C., & Bradford, W. D. (2020). The effect of evictions on accidental drug and alcohol mortality. *Health Services Research*, *55*(1), 9–17. https://doi.org/10.1111/1475-6773.13256

Case, A., & Deaton, A. (2022). The great divide: Education, despair, and death. *Annual Review of Economics*, *14*, 1–21. https://doi.org/10.1146/annurev-economics-051520-015607

Case, A., & Deaton, A. (2015). Rising morbidity and mortality in midlife among white non-hispanic americans in the 21st century. *Proceedings of the National Academy of Sciences*, *112*(49), 15078–15083. https://doi.org/10.1073/pnas.1518393112

Case, A., & Deaton, A. (2020). *Deaths of despair and the future of capitalism*. Princeton University Press.

Ciminelli, G., Duval, R., & Furceri, D. (2022). Employment protection deregulation and labor shares in advanced economies. *Review of Economics and Statistics*, *104*(6), 1174–1190.

Darke, S. (2010). The toxicology of homicide offenders and victims: A review. *Drug and Alcohol Review*, *29*(2), 202–215. https://doi.org/10.1111/j.1465-3362.2009.00145.x

Dow, W. H., Godøy, A., Lowenstein, C. A., & Reich, M. (2020). Can labor-market policies reduce deaths of despair? *Journal of Health Economics*, *74*, 102372. https://doi.org/10.1016/j.jhealeco.2020.102372

Dube, A., Girardi, D., Jordà, Ò., & Taylor, A. M. (2025). A local projections approach to difference-in-differences. *Journal of Applied Econometrics*, *40*(7). https://doi.org/10.1002/jae.70000

Fishman, S. H., & Gutin, I. (2021). Debts of despair: Education, financial losses, and precursors of deaths of despair. *SSM–Population Health*, *14*, 100759. https://doi.org/10.1016/j.ssmph.2021.100759

Foran, H. M., & O'Leary, K. D. (2008). Alcohol and intimate partner violence: A meta-analytic review. *Clinical Psychology Review*, *28*(7), 1222–1234. https://doi.org/10.1016/j.cpr.2008.05.001

Gaydosh, L., Hummer, R. A., Hargrove, T. W., Halpern, C. T., Hussey, J. M., Whitsel, E. A., Dole, N., & Harris, K. M. (2019). The depths of despair among us adults entering midlife. *American Journal of Public Health*, *109*(5), 774–780. https://doi.org/10.2105/AJPH.2019.305002

Geronimus, A. T., Bound, J., Waidmann, T. A., Rodriguez, J. M., & Timpe, B. (2019). Weathering, drugs, and whack-a-mole: Fundamental and proximate causes of widening educational inequity in u.s. life expectancy by sex and race, 1990–2015. *Journal of Health and Social Behavior*, *60*(2), 222–239. https://doi.org/10.1177/0022146519849932

Glei, D. A., Lee, C., Brown, C. L., & Weinstein, M. (2024). Are employment and social integration more strongly associated with deaths of despair than psychological or economic distress? *Social Science & Medicine*, *357*, 117197. https://doi.org/10.1016/j.socscimed.2024.117197

Gutin, I., & Hummer, R. A. (2020). Occupation, employment status, and "despair"-associated mortality risk among working-aged u.s. adults, 1997–2015. *Preventive Medicine*, *137*, 106129. https://doi.org/10.1016/j.ypmed.2020.106129

Hammarström, A., Westerlund, H., Janlert, U., Virtanen, P., Ziaei, S., & Östergren, P.-O. (2024). How do labour market conditions explain the development of mental health over the life-course? a conceptual integration of the ecological model with life-course epidemiology in an integrative review of results from the northern swedish cohort. *BMC Public Health*, *24*(1), 1315. https://doi.org/10.1186/s12889-024-18461-6

Hawkins, D., Punnett, L., Davis, L., & Kriebel, D. (2021). The contribution of occupation-specific factors to the deaths of despair, massachusetts, 2005–2015. *Annals of Work Exposures and Health*, *65*(7), 819–832. https://doi.org/10.1093/annweh/wxab017

Jolivet, G., & Postel-Vinay, F. (2025). A structural analysis of mental health and labour market trajectories. *Review of Economic Studies*, *92*(3), 1920–1954.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, *95*(1), 161–182.

Jou, A., Mas, N., & Vergara-Alert, C. (2023). Housing wealth, health and deaths of despair. *Journal of Real Estate Finance and Economics*, *66*(3), 569–602. https://doi.org/10.1007/s11146-020-09801-5

Knapp, E. A., Bilal, U., Dean, L. T., Lazo, M., & Celentano, D. D. (2019). Economic insecurity and deaths of despair in us counties. *American Journal of Epidemiology*, *188*(12), 2131–2139.

Kuhns, J. B., Exum, M. L., Clodfelter, T. A., & Bottia, M. C. (2014). The prevalence of alcohol-involved homicide offending: A meta-analytic review. *Homicide Studies*, *18*(3), 251–270. https://doi.org/10.1177/1088767914522524

Kuo, C.-T., & Kawachi, I. (2023). County-level income inequality, social mobility, and deaths of despair in the us, 2000–2019. *JAMA Network Open*, *6*(7), e2323030. https://doi.org/10.1001/jamanetworkopen.2023.23030

Link, B. G., & Phelan, J. (1995). Social conditions as fundamental causes of disease. *Journal of Health and Social Behavior*, 80–94.

Mishra, A., Nafuna, S., & Abi, E. A. (2024). Do robots cause deaths of despair? [Working paper, SSRN preprint]. https://ssrn.com/abstract=4762118

Monnat, S. M. (2020). Trends in u.s. working-age non-hispanic white mortality: Rural–urban and within-rural differences. *Population Research and Policy Review*, *39*(5), 805–834. https://doi.org/10.1007/s11113-020-09607-6

OECD. (2023). *Health at a glance 2023: Oecd indicators*. https://doi.org/10.1787/7a7afb35-en

Peters, J. (2008). Labour market deregulation and the decline of labour power in north america and western europe. *Policy and Society*, *27*(1), 83–98. https://doi.org/10.1016/j.polsoc.2008.07.007

Piano, M. R., & Phillips, S. A. (2023). Alcoholic cardiomyopathy: Pathophysiology and clinical management. *Journal of Cardiac Failure*, *29*(4), 567–578. https://doi.org/10.1016/j.cardfail.2023.01.005

Pierce, J. R., & Schott, P. K. (2020). Trade liberalization and mortality: Evidence from u.s. counties. *American Economic Review: Insights*, *2*(1), 47–64. https://doi.org/10.1257/aeri.20180396

Putnam, R. D. (2000). *Bowling alone: The collapse and revival of american community*. Simon; Schuster.

Ruhm, C. J. (2022). Living and dying in america: An essay on deaths of despair and the future of capitalism [Originally issued as NBER Working Paper No. 28358 (2021)]. *Journal of Economic Literature*.

Ruhm, C. J. (2018). *Deaths of despair or drug problems?* (NBER Working Paper No. 24188). National Bureau of Economic Research. Cambridge, MA. http://www.nber.org/papers/w24188

Scheiring, G., Stuckler, D., & King, L. (2020). *Deindustrialization and deaths of despair: Mapping the impact of industrial decline on ill health* (tech. rep. No. PERI Working Paper No. 530). Political Economy Research Institute, University of Massachusetts Amherst. https://doi.org/10.7275/27137530

Schmidt, M., Habib, G., & Hoen, B. (2022). Infective endocarditis in people who inject drugs: Epidemiology and outcomes. *Clinical Infectious Diseases*, *75*(3), e123–e131. https://doi.org/10.1093/cid/ciac123

Stein, E. M., Gennuso, K. P., Ugboaja, D. C., & Remington, P. L. (2017). The epidemic of despair among white americans: Trends in the leading causes of premature death, 1999–2015. *American Journal of Public Health*, *107*(10), 1541–1547. https://doi.org/10.2105/AJPH.2017.303941

U.S. Census Bureau. (2025, September). Census bureau releases new educational attainment data [Accessed: 2026-06-02]. https://census.gov

Venkataramani, A. S., & Tsai, A. C. (2020). Housing, housing policy, and deaths of despair. *Health Services Research*, *55*(1), 5–8. https://doi.org/10.1111/1475-6773.13260

Virmani, R., Kolodgie, F. D., Burke, A. P., Farb, A., & Schwartz, S. M. (2001). Lessons from sudden coronary death: A comprehensive morphological classification scheme for atherosclerotic lesions. *Circulation*, *103*(7), 953–959. https://doi.org/10.1161/01.CIR.103.7.953

Western, B., & Rosenfeld, J. (2011). Unions, norms, and the rise in u.s. wage inequality. *American Sociological Review*, *76*(4), 513–537. https://doi.org/10.1177/0003122411414817

Wilkinson, R. G., & Marmot, M. (Eds.). (2003). *Social determinants of health: The solid facts*. World Health Organization.

World Health Organization. (2018). *Global status report on alcohol and health 2018*. WHO. https://www.who.int/publications/i/item/9789241565639

9 Appendix

9.1 Data Appendix

Table 5: ICD-10 Codes and Descriptions Grouped by Cause of Death

Cause of Death	ICD-10 Code	Description
Drug Use Disorders	F11	Mental and behavioral disorders due to use of opioids
	F12	Mental and behavioral disorders due to use of cannabinoids
	F13	Mental and behavioral disorders due to use of sedatives or hypnotics
	F14	Mental and behavioral disorders due to use of cocaine
	F15	Mental and behavioral disorders due to use of other stimulants, including caffeine
	F16	Mental and behavioral disorders due to use of hallucinogens
	F18	Mental and behavioral disorders due to use of volatile solvents
	F19	Mental and behavioral disorders due to multiple drug use and other psychoactive substances
	X41	Accidental poisoning by antiepileptic, sedative-hypnotic, antiparkinsonism and psychotropic drugs
	X42	Accidental poisoning by narcotics and hallucinogens
	X44	Accidental poisoning by other and unspecified drugs
Alcohol Use Disorders	F10	Mental and behavioral disorders due to use of alcohol
	X45	Accidental poisoning by and exposure to alcohol
Self-Inflicted Injuries	X60–X84	Intentional self-harm
	Y870	Sequelae of intentional self-harm
Violence	X85–Y09	Assault
	Y871	Sequelae of assault
Inflammatory Heart Diseases	I30	Acute pericarditis
	I31	Other diseases of pericardium
	I32	Pericarditis in diseases classified elsewhere
	I33	Acute and subacute endocarditis
	I38	Endocarditis, valve unspecified
	I40	Acute myocarditis
	I42	Cardiomyopathy

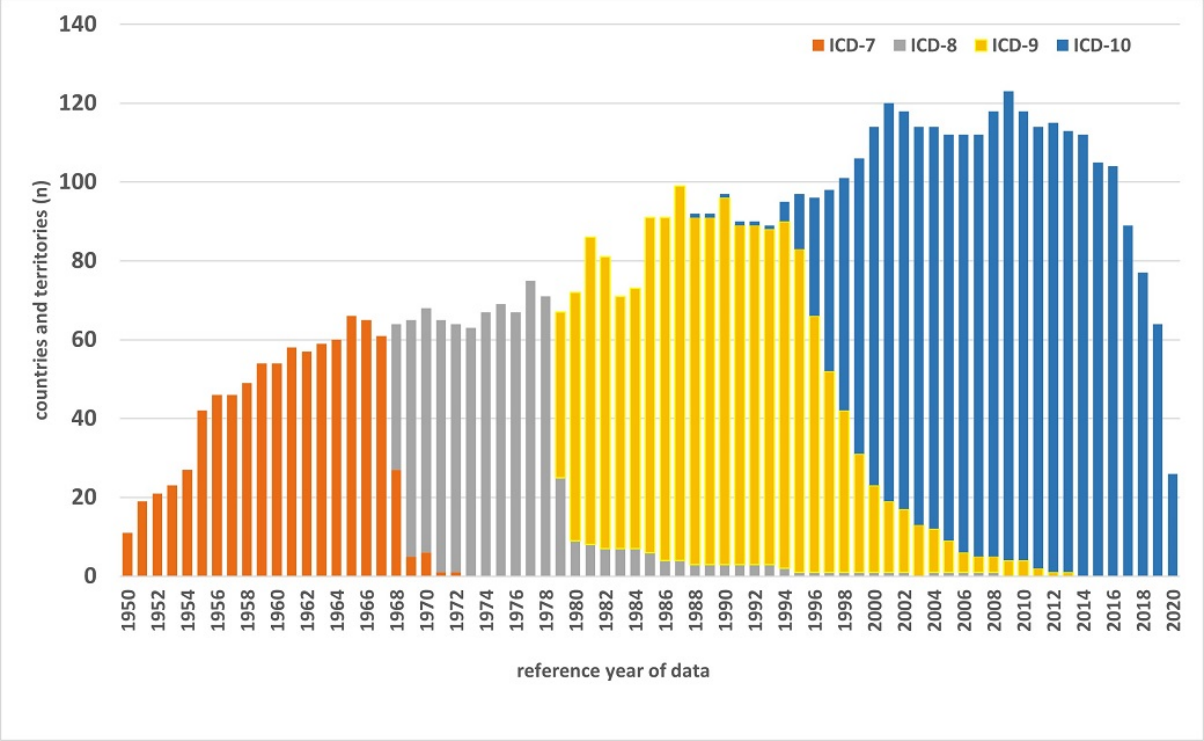
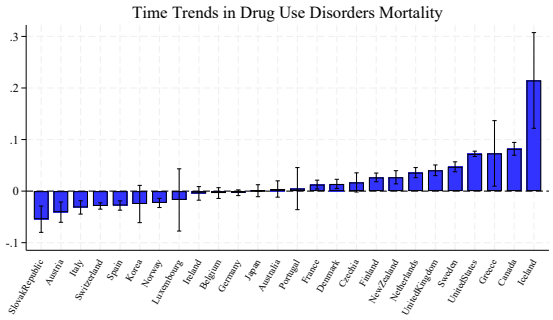
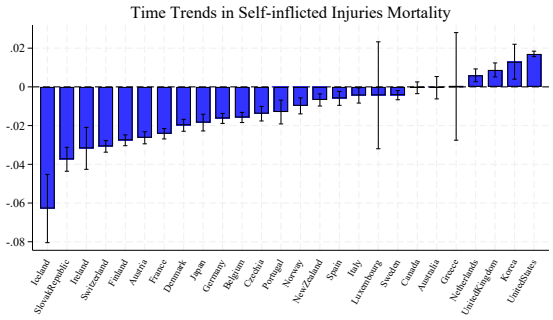


Figure 11: Number of Countries in ICD Transition

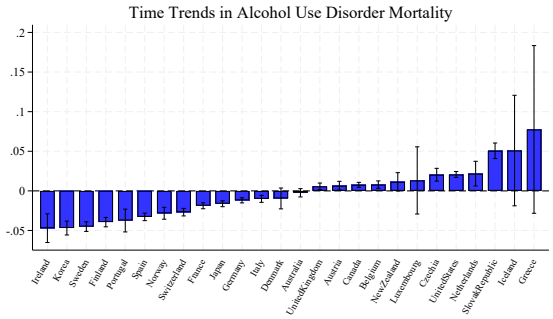
9.2 Descriptive Statistics: Country-Specific Trends



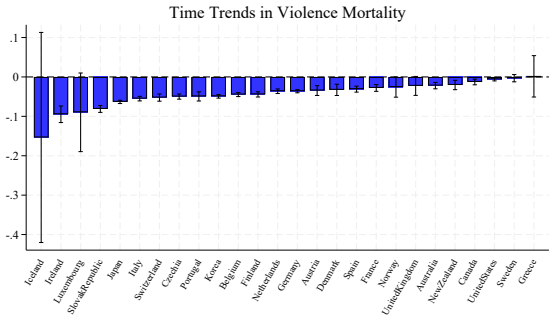
(a) ln(Drug)



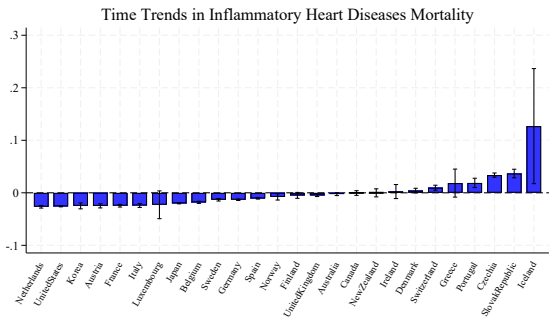
(b) ln(Suicide)



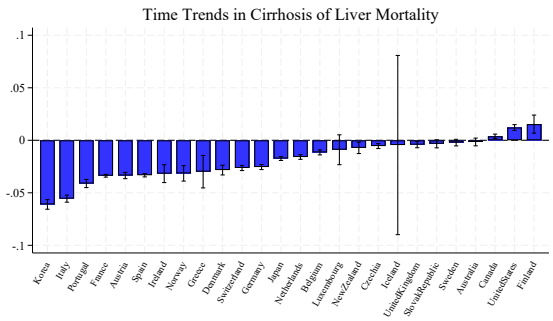
(c) ln(Alcohol)



(d) ln(Violence)



(e) ln(Inflammatory Heart Disease)



(f) ln(Cirrhosis of Liver)

Figure 12: **Country-specific time trends in log mortality rates by cause.** Each panel shows estimated linear trends in log age-adjusted mortality for the indicated cause of death of each country. Trends are obtained from the regression $Mortality_t^i = \alpha^i + \tau^i t + \varepsilon_t^i$, where the subscript t and superscript i denote year and country, respectively. $Mortality$ denotes the cause-specific mortality rates, α^i is a country-specific constant, τ^i is the country-specific linear time trend, and ε_t^i is an error term. Capped spikes indicate 90 percent confidence intervals. The estimates can be interpreted as the average annual change in a cause-specific mortality rate for each country over the sample period, which for most countries spans 1998–2019.

9.3 Additional Heterogeneity Results by Institutional Context

Table 6: Dynamic effects of EPL deregulation on mortality from self-inflicted injury: interaction terms

Interaction term with	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Avoidable death rate	-0.0001	-0.02	-0.01	-0.03	-0.01	-0.02
	(0.0177)	(0.0306)	(0.0301)	(0.0528)	(0.0695)	(0.0789)
Social transfers expenditure	0.04	0.09**	0.06	0.10	0.16**	0.18**
	(0.0286)	(0.0403)	(0.0506)	(0.0763)	(0.0662)	(0.0782)
Public health insurance coverage	-0.0018	-0.0049**	-0.0031	-0.0039	-0.0038	0.0008
	(0.0011)	(0.0020)	(0.0030)	(0.0051)	(0.0129)	(0.0184)
Replacement rate of UB (60 months)	0.02	0.04	0.02	-0.01	-0.01	0.06
	(0.0187)	(0.0324)	(0.0397)	(0.0467)	(0.0594)	(0.0582)
Poverty rate	0.01	0.00	-0.04	-0.01	-0.04	-0.01
	(0.0320)	(0.0570)	(0.0675)	(0.0691)	(0.0790)	(0.0557)

Notes: Outcome is the **log mortality rate from self-inflicted injury (suicide)** per 100,000 population. Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: $^{*}p < 0.10$, $^{*}\hat{p} < 0.05$, $^{*}\hat{p}^{*} < 0.01$. If figures report percent changes, coefficients can be transformed as $100\,[\exp(\hat{\beta}_h) - 1]\%$.

Table 7: Dynamic effects of EPL deregulation on mortality from alcohol use disorder: interaction terms

Interaction term with	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Avoidable death rate	-0.02	0.03	0.03	-0.01	0.04	0.11
	(0.0379)	(0.0540)	(0.0743)	(0.0955)	(0.1192)	(0.1170)
Social transfers expenditure	-0.18***	-0.21**	-0.12	-0.28***	-0.33***	-0.24
	(0.0411)	(0.0832)	(0.0971)	(0.0741)	(0.0985)	(0.1420)
Public health insurance coverage	-0.01	0.03***	0.02***	0.03*	0.001	-0.003
	(0.0083)	(0.0064)	(0.0061)	(0.0158)	(0.0162)	(0.0261)
Replacement rate of UB (60 months)	-0.06	-0.12*	-0.13*	-0.10	-0.04	-0.01
	(0.0448)	(0.0639)	(0.0717)	(0.0794)	(0.1417)	(0.0979)
Poverty rate	-0.12***	-0.09	0.10	0.07	0.15	-0.10
	(0.0387)	(0.0488)	(0.0617)	(0.0704)	(0.1671)	(0.0969)

Notes: Outcome is the **log mortality rate from alcohol use disorder** (per 100,000 population). Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: $^{*}p < 0.10$, $^{*}\hat{p} < 0.05$, $^{*}\hat{p}^{*} < 0.01$. If figures report percent changes, coefficients can be transformed as $100\,[\exp(\hat{\beta}_h) - 1]\%$.

Table 8: Dynamic effects of EPL deregulation on mortality from violence: interaction terms

Interaction term with	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Avoidable death rate	0.08*	0.09	0.07	0.03	-0.13	0.04
	(0.0395)	(0.0756)	(0.0762)	(0.0846)	(0.1000)	(0.1450)
Social transfers expenditure	-0.02	0.03	-0.14	-0.12	-0.21	0.07
	(0.0612)	(0.0613)	(0.1603)	(0.1534)	(0.2052)	(0.2606)
Public health insurance coverage	0.01	-0.01	0.01	-0.01	-0.03	-0.06
	(0.0045)	(0.0056)	(0.0107)	(0.0167)	(0.0339)	(0.0359)
Replacement rate of UB (60 months)	0.06	0.12	0.07	0.28**	0.39***	0.54***
	(0.0684)	(0.0831)	(0.1137)	(0.0984)	(0.1204)	(0.1166)
Poverty rate	-0.04	-0.07	-0.29**	-0.10	-0.07	-0.20
	(0.0928)	(0.1088)	(0.1043)	(0.1484)	(0.1439)	(0.1843)

Notes: Outcome is the **log mortality rate from violence** (per 100,000 population). Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: $^{*}p < 0.10$, $^{*}\hat{p} < 0.05$, $^{*}\hat{p}^{*} < 0.01$. Coefficients can be transformed to percent changes using $100[\exp(\hat{\beta}_h) - 1]\%$.

Table 9: Dynamic effects of EPL deregulation on mortality from inflammatory heart diseases: interaction terms

Interaction term with	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Avoidable death rate	-0.004	0.003	0.02	0.07***	0.08**	0.07**
	(0.0159)	(0.0197)	(0.0175)	(0.0203)	(0.0287)	(0.0266)
Social transfers expenditure	-0.06**	-0.02	-0.04	0.02	-0.02	0.02
	(0.0234)	(0.0317)	(0.0539)	(0.0490)	(0.0721)	(0.0469)
Public health insurance coverage	-0.002***	0.001	-0.005	-0.001	-0.02**	-0.03*
	(0.0004)	(0.0010)	(0.0027)	(0.0043)	(0.0098)	(0.0177)
Replacement rate of UB (60 months)	-0.02	-0.01	0.01	0.002	0.0002	-0.02
	(0.0286)	(0.0416)	(0.0572)	(0.0537)	(0.0602)	(0.0585)
Poverty rate	-0.05	-0.06	-0.10**	-0.19**	-0.42***	-0.30***
	(0.0318)	(0.0603)	(0.0427)	(0.0658)	(0.0351)	(0.0551)

Notes: Outcome is the **log mortality rate from inflammatory heart diseases** (per 100,000 population). Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$. Percent-change interpretation (if used in figures): $100[\exp(\hat{\beta}_h) - 1]\%$.

9.4 Asymmetry Between Deregulation and Tightening

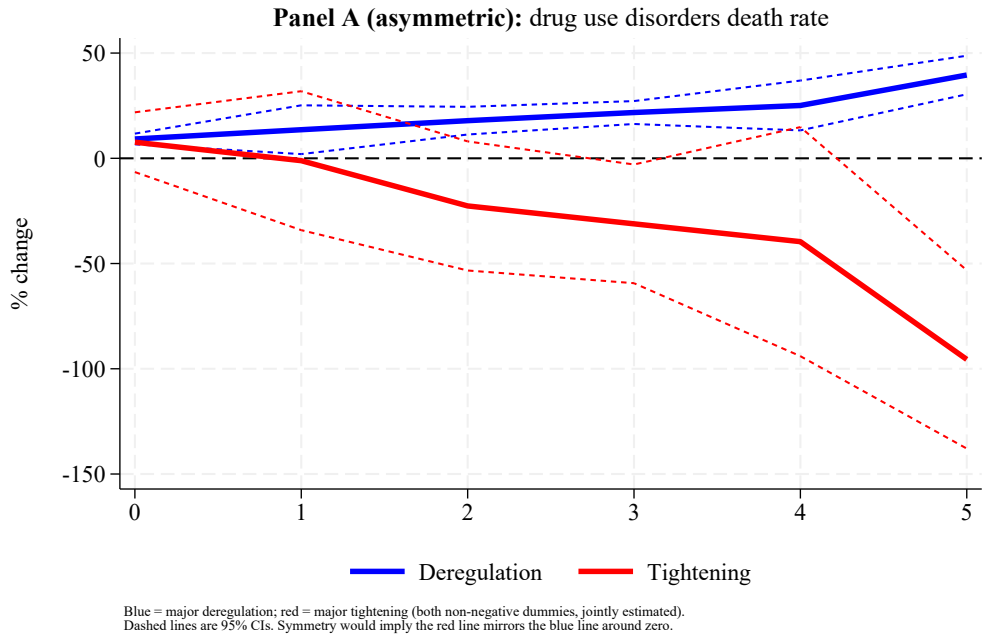


Figure 13: **Asymmetry between EPL deregulation and tightening: drug use disorder mortality.** The signed reform indicator used in the baseline specification is decomposed into two non-negative dummies, one for major deregulation episodes and one for major tightening episodes, entered jointly with their own lags and forward-term corrections. The blue line plots the deregulation coefficient (nearly identical to the baseline estimates in Table 2); the red line plots the tightening coefficient. Dashed lines are 95% confidence intervals. Under a symmetric response—the restriction imposed by construction in the baseline specification—the red line would mirror the blue line around zero. Coefficients and the corresponding horizon-by-horizon test of that restriction are reported in Table 10.

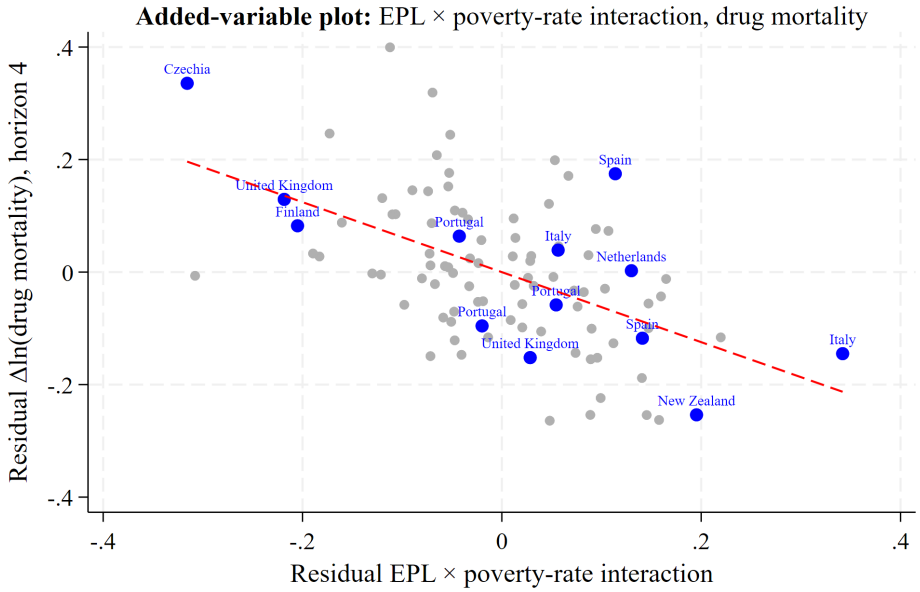
Table 10: Asymmetry between EPL deregulation and tightening: drug use disorder mortality

Horizon	**0y**	**1y**	**2y**	**3y**	**4y**	**5y**
Deregulation, β_{dereg}	**0.092**	**0.136**	**0.179**	**0.218**	**0.251**	**0.396**
	[0.066, 0.117]	[0.020, 0.252]	[0.113, 0.245]	[0.163, 0.272]	[0.133, 0.369]	[0.304, 0.487]
Tightening, $\beta_{tighten}$	0.077	−0.011	−0.226	**−0.311**	−0.396	**−0.955**
	[−0.065, 0.219]	[−0.341, 0.319]	[−0.533, 0.080]	[−0.593, −0.029]	[−0.940, 0.148]	[−1.379, −0.532]
Symmetry test p**-value**[a]	0.020**	0.466	0.761	0.517	0.608	0.016**

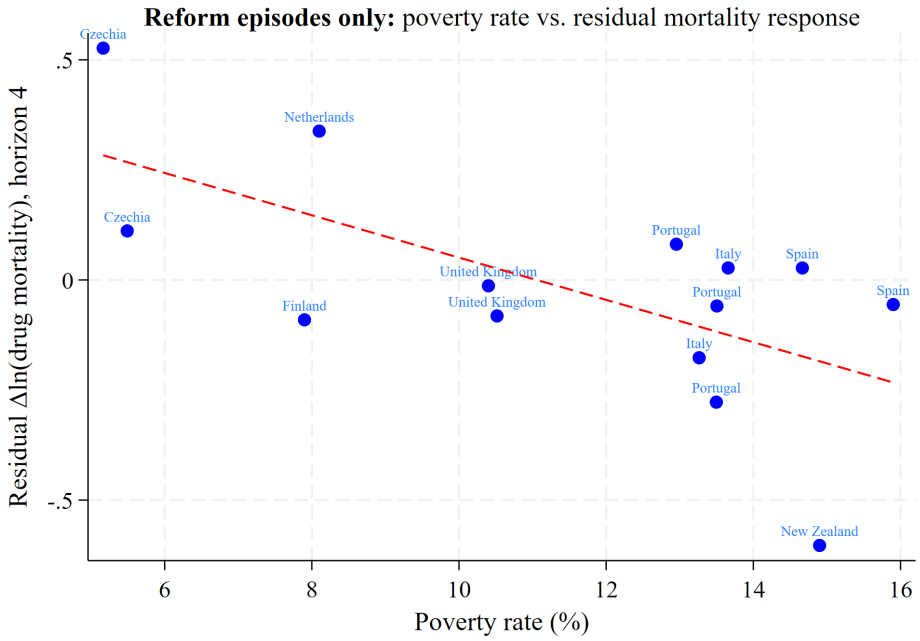
Notes: Outcome is the log mortality rate from drug use disorders (per 100,000 population). β_{dereg} and $\beta_{tighten}$ are the coefficients on non-negative dummies for major deregulation and major tightening episodes, respectively, jointly estimated in place of the signed `epl_flag` indicator used in the baseline specification, with their own lags and horizon-specific forward-term corrections; 95% confidence intervals in brackets. Bold coefficients indicate a 95% confidence interval that excludes zero.

[a] Tests H_0: $\beta_{dereg} + \beta_{tighten} = 0$, i.e., that tightening reverses the mortality effect of deregulation exactly—the restriction imposed by construction in the baseline symmetric specification. *$p < 0.10$, **$p < 0.05$.

9.5 Additional Robustness Checks



(a) Added-variable plot



(b) Poverty rate vs. residual response, reform episodes only

Figure 14: **Is the EPL × poverty-rate interaction driven by outliers?** Panel (a) plots the residual of the horizon-3 drug mortality response against the residual of the EPL × poverty-rate interaction term, after partialling out all other regressors in the specification (country and year fixed effects, controls, lags, and forward reform terms); by the Frisch–Waugh–Lovell theorem, the slope of the fitted line reproduces the interaction coefficient reported in Table 3. Blue markers denote country-years with a major EPL reform, labeled by country (repeated country names indicate multiple reform episodes for that country); gray markers denote non-reform country-years. Panel (b) plots the same residualized mortality response against the raw poverty rate, restricted to reform episodes only. Both panels indicate that the negative interaction reflects a pattern shared across roughly a dozen reform episodes spanning the poverty distribution, rather than one or two extreme observations.

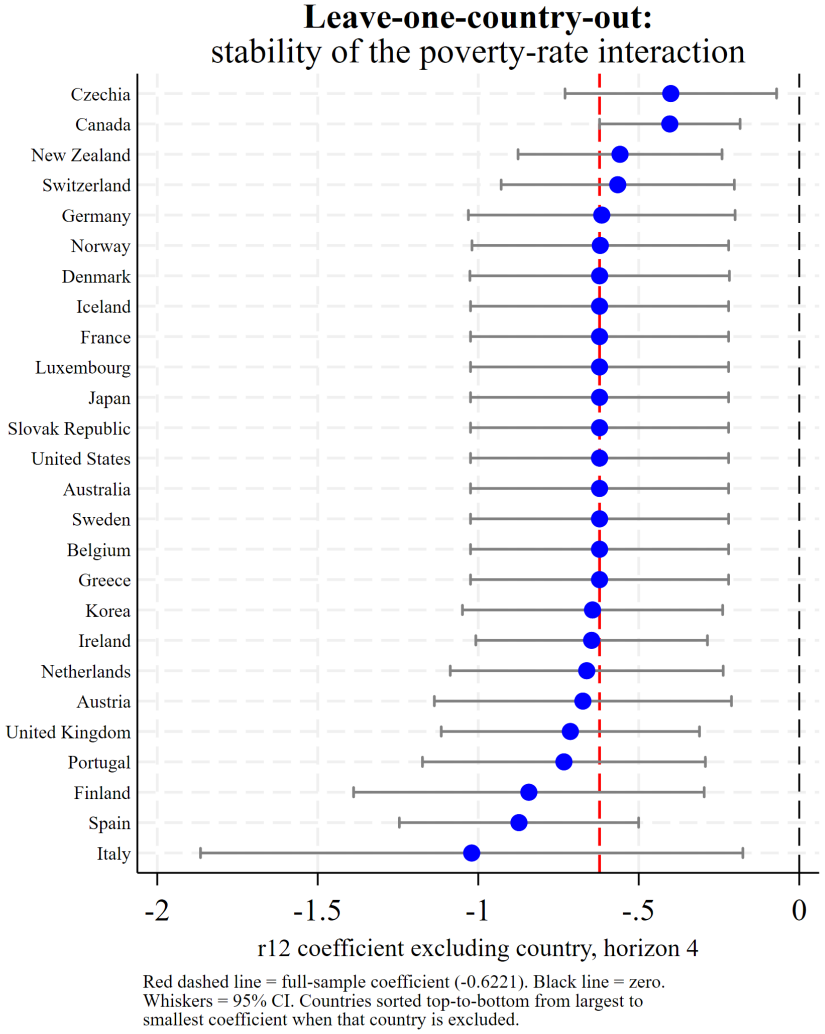


Figure 15: **Leave-one-country-out estimates of the EPL × poverty-rate interaction.** Each row re-estimates the horizon-3 interaction coefficient from Table 3 after excluding the named country from the sample. Whiskers are 95% confidence intervals; the red dashed line marks the full-sample coefficient (-0.622). The estimated coefficient remains negative, with a confidence interval that excludes zero, regardless of which single country is dropped, ranging from -0.35 (excluding Czechia) to -1.05 (excluding Italy).

Table 11: Dynamic effects of EPL deregulation on mortality from drug use disorders by age group

Age group	0y	1y	2y	3y	4y	5y
15–19	0.03	-0.03	-0.09	-0.05	-0.08	-0.55***
	(0.1147)	(0.1774)	(0.2286)	(0.1927)	(0.1464)	(0.1176)
20–24	0.00	0.14***	0.17*	0.40***	0.16	0.36***
	(0.1006)	(0.0375)	(0.0949)	(0.0700)	(0.1140)	(0.0668)
25–29	-0.07	0.07	0.38***	0.22**	0.35***	0.50***
	(0.1093)	(0.0665)	(0.0874)	(0.0825)	(0.0510)	(0.1179)
30–34	0.08	0.12	0.31***	0.16	0.44***	0.42***
	(0.0935)	(0.1545)	(0.0557)	(0.0904)	(0.1067)	(0.0691)
35–39	0.04	0.11***	0.23***	0.20***	0.32***	0.29***
	(0.0456)	(0.0341)	(0.0657)	(0.0557)	(0.0700)	(0.0500)
40–44	0.01	-0.10	0.12	0.17**	0.22***	0.26***
	(0.0614)	(0.0824)	(0.0778)	(0.0565)	(0.0533)	(0.0615)
45–49	0.17**	0.12**	0.17**	0.11	0.07	0.25***
	(0.0668)	(0.0550)	(0.0604)	(0.0722)	(0.0982)	(0.0631)
50–54	-0.04	0.22**	0.13**	0.19	0.23***	0.48***
	(0.1236)	(0.0915)	(0.0538)	(0.1333)	(0.0531)	(0.0554)
55–59	0.01	0.11	0.08	0.15*	0.13*	0.34**
	(0.1226)	(0.0861)	(0.0963)	(0.0787)	(0.0704)	(0.1245)
60–64	0.19*	0.24**	0.35***	0.40**	0.27***	0.41***
	(0.0973)	(0.0961)	(0.0750)	(0.1317)	(0.0749)	(0.1029)
65–69	0.07	0.25*	0.12	0.32***	0.20***	0.38***
	(0.0961)	(0.1327)	(0.1489)	(0.0937)	(0.0641)	(0.0893)
70–74	-0.30**	0.06	-0.05	-0.24***	0.13	-0.17
	(0.1207)	(0.1104)	(0.0935)	(0.0772)	(0.1118)	(0.1126)
75–79	-0.10	0.11	0.14	0.06	0.16	0.14
	(0.1498)	(0.0986)	(0.0913)	(0.0879)	(0.1804)	(0.1384)

Notes: Outcome is the **log mortality rate from drug use disorders** (per 100,000 population). Entries are age-group–specific dynamic effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: $^{*}p < 0.10$, $^{*}p < 0.05$, $^{*}p < 0.01$. For percent-change interpretation (used in figures if applicable), transform as $100[\exp(\hat{\beta}_h) - 1]\%$.